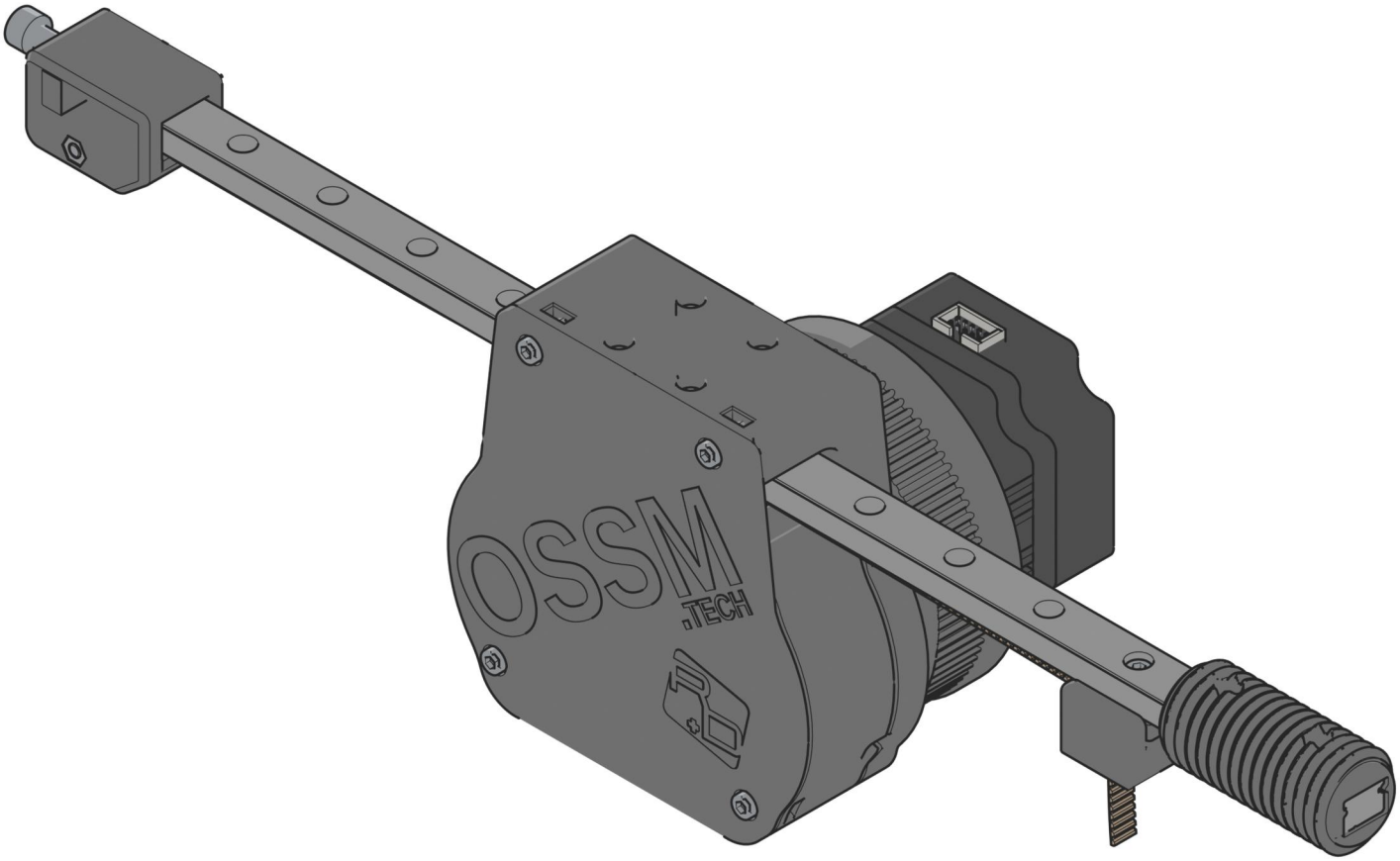
OSSM

RESEARCH + DESIRE

Actuator Build Guide

Actuator · Version 1.0.15



Support
support@researchanddesire.com

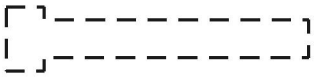
Assembly instructions
ohai.researchanddesire.com/ossm

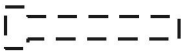
Actual size

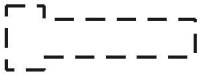


Print at 100 % (actual size). Lay a part on its dashed outline to identify it.

Bolts


M5×35 Screw


M3×20 Screw


M5×20 Screw


M3×8 Screw

Nuts


M5 Nut


M3 Nut

Hex keys

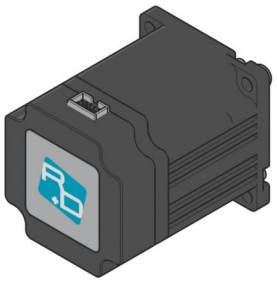

4 mm Hex key


2.5 mm Hex key

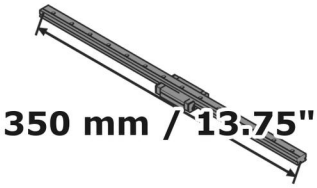

2 mm Hex key

Parts

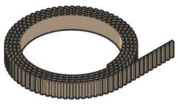
Motor and linear rail



1×
Servo motor

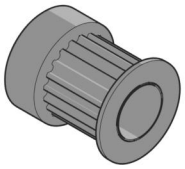


1×
Linear rail



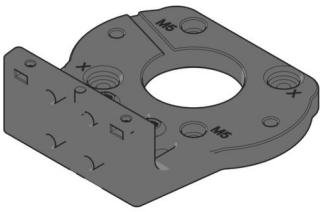
500 mm

1×
Belt

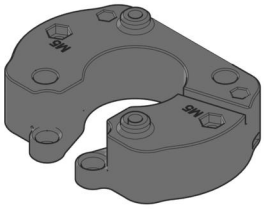


10 mm
1×
Pulley

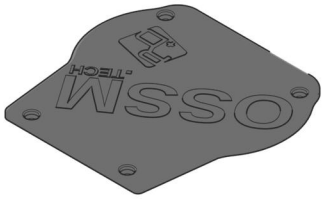
3D printed



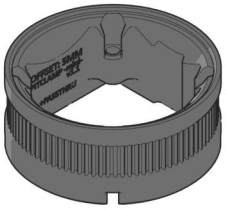
1×
Body bottom



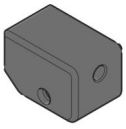
1×
Body middle



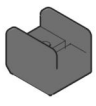
1×
Cover



1×
Motor ring



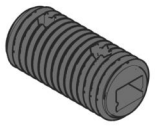
1×
Belt tensioner



1×
Belt clamp bottom



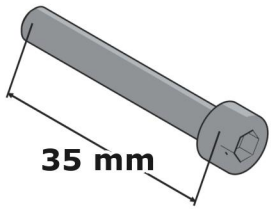
1×
Belt clamp top



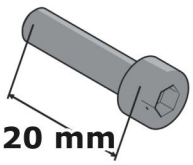
1×
24 mm
Threaded end effector

Parts

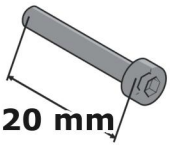
Bolts



4×
M5×35 Screw



3×
M5×20 Screw



2×
M3×20 Screw



10×
M3×8 Screw

Nuts



5×
M5 Nut



8×
M3 Nut

Bearings



6×
5×11×4
Bearing

Tools

Required



4 mm Hex key

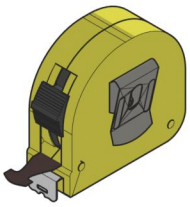


2.5 mm
Hex key



2 mm Hex key

Recommended



Tape measure

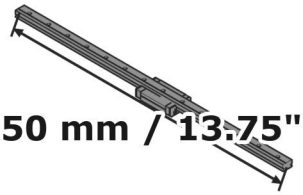


Threadlocker
(blue 243)

Legend

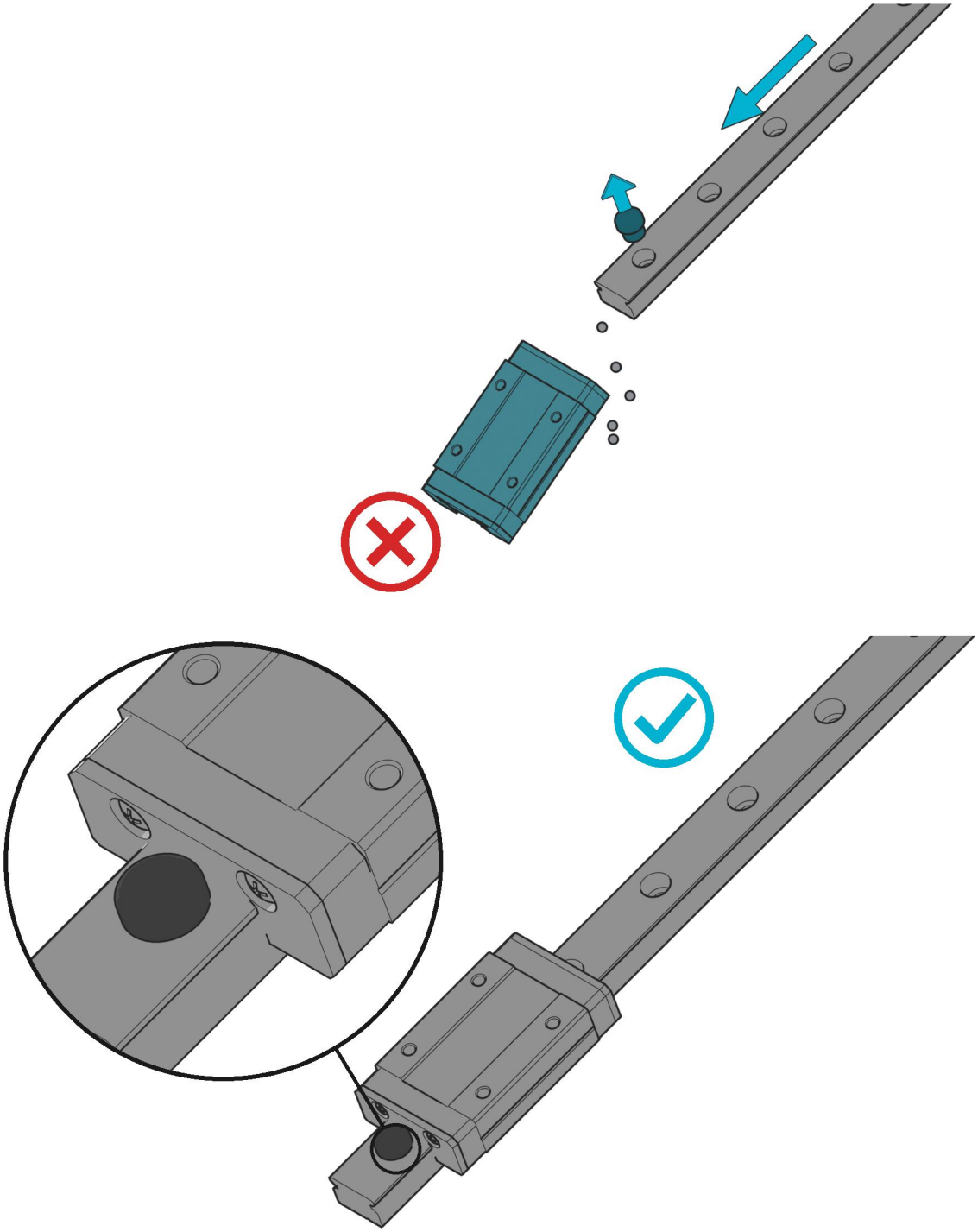
In a step's parts box, this drop beside a screw's quantity means:
put threadlocker (blue 243) on the screw's thread before it goes in.

Warnings ⚠



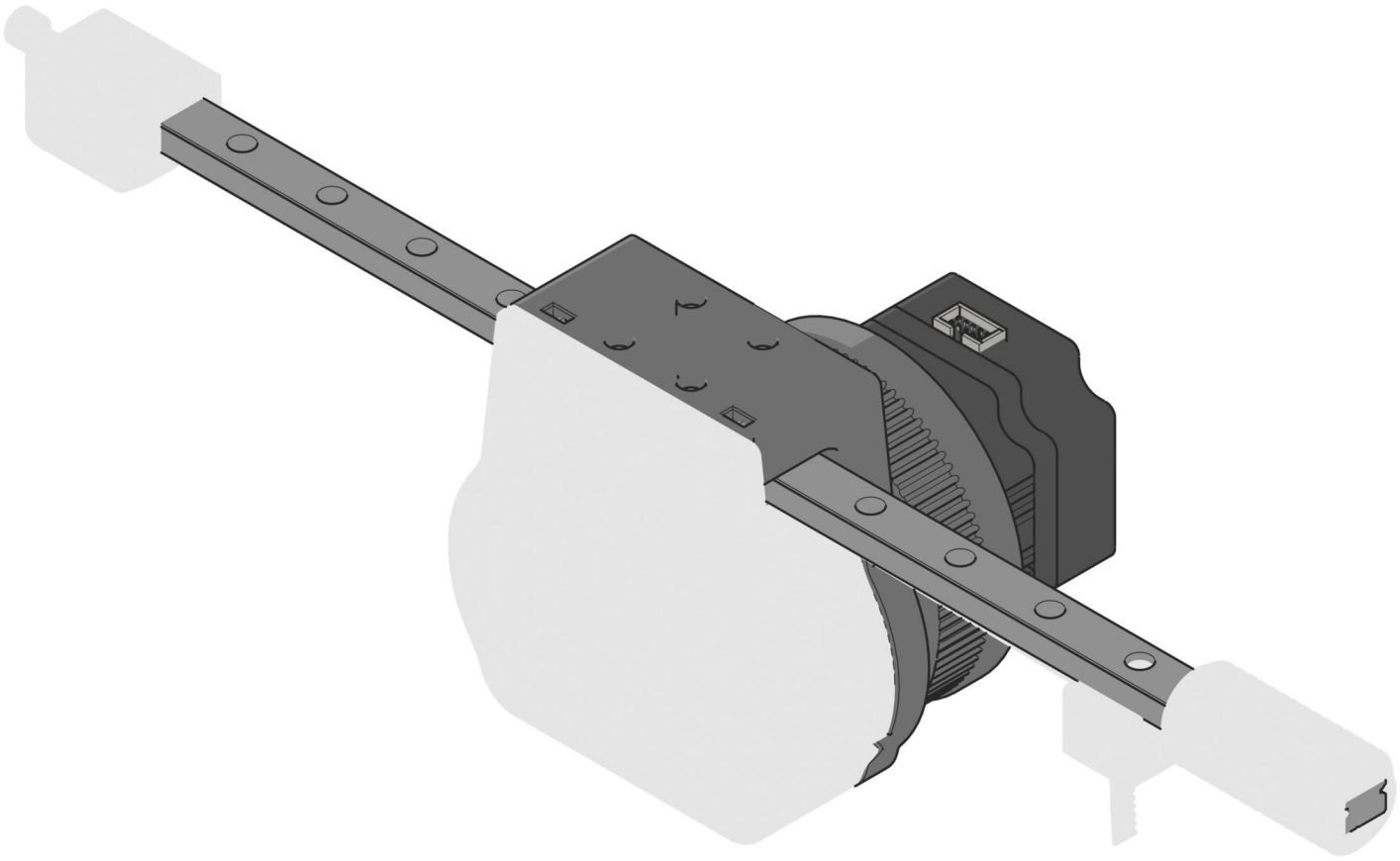
1×

Linear rail



⚠ **Never let the carriage slide off the rail: its balls fall out.**

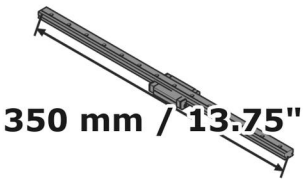
1 The motor head



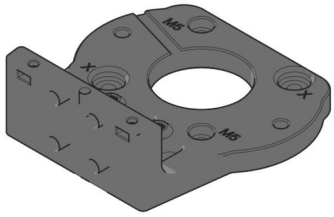
1



4×
M3×8 Screw



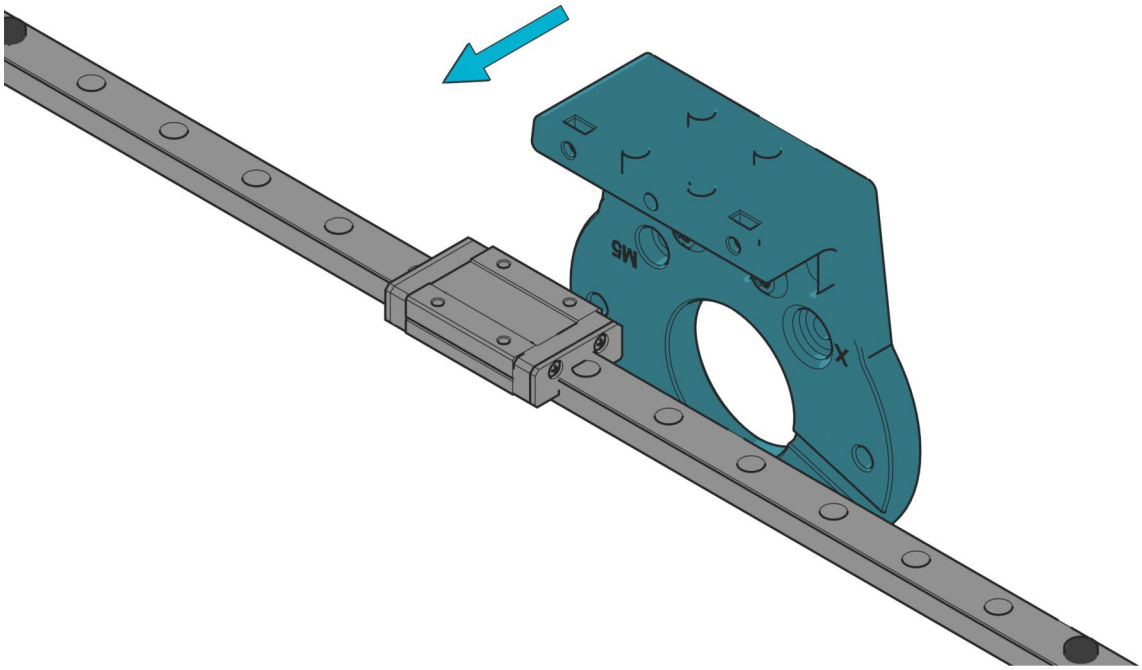
1×
Linear rail

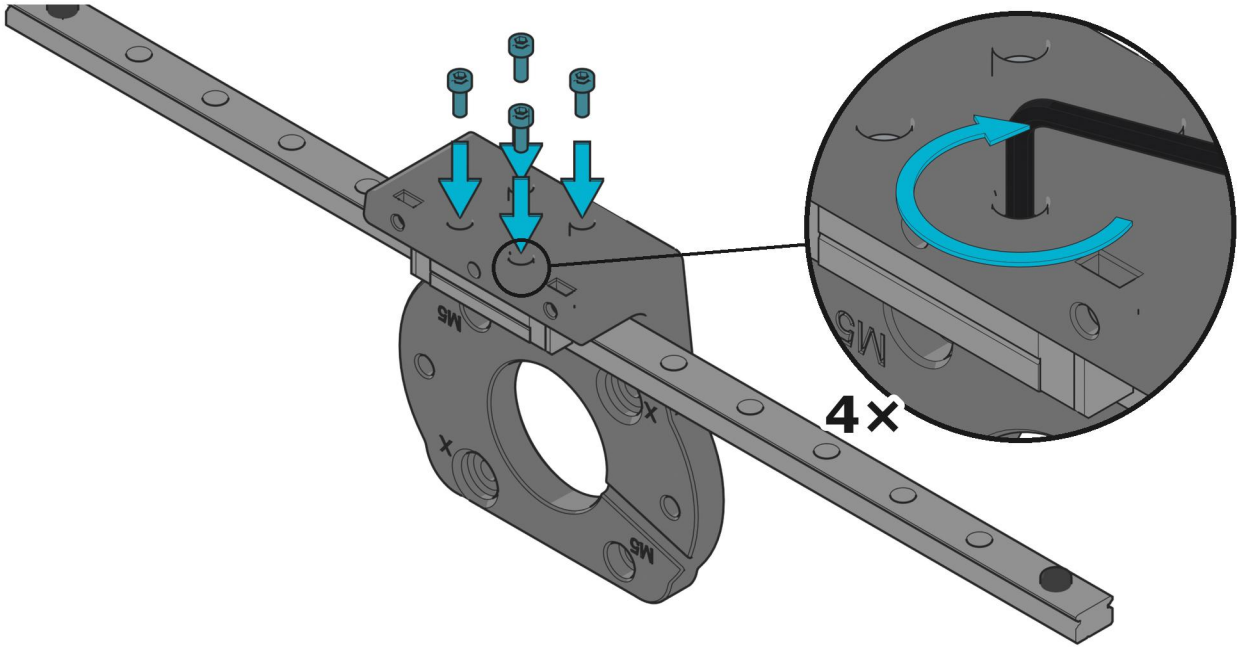


1×
Body bottom

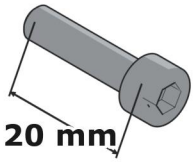


2.5 mm Hex key



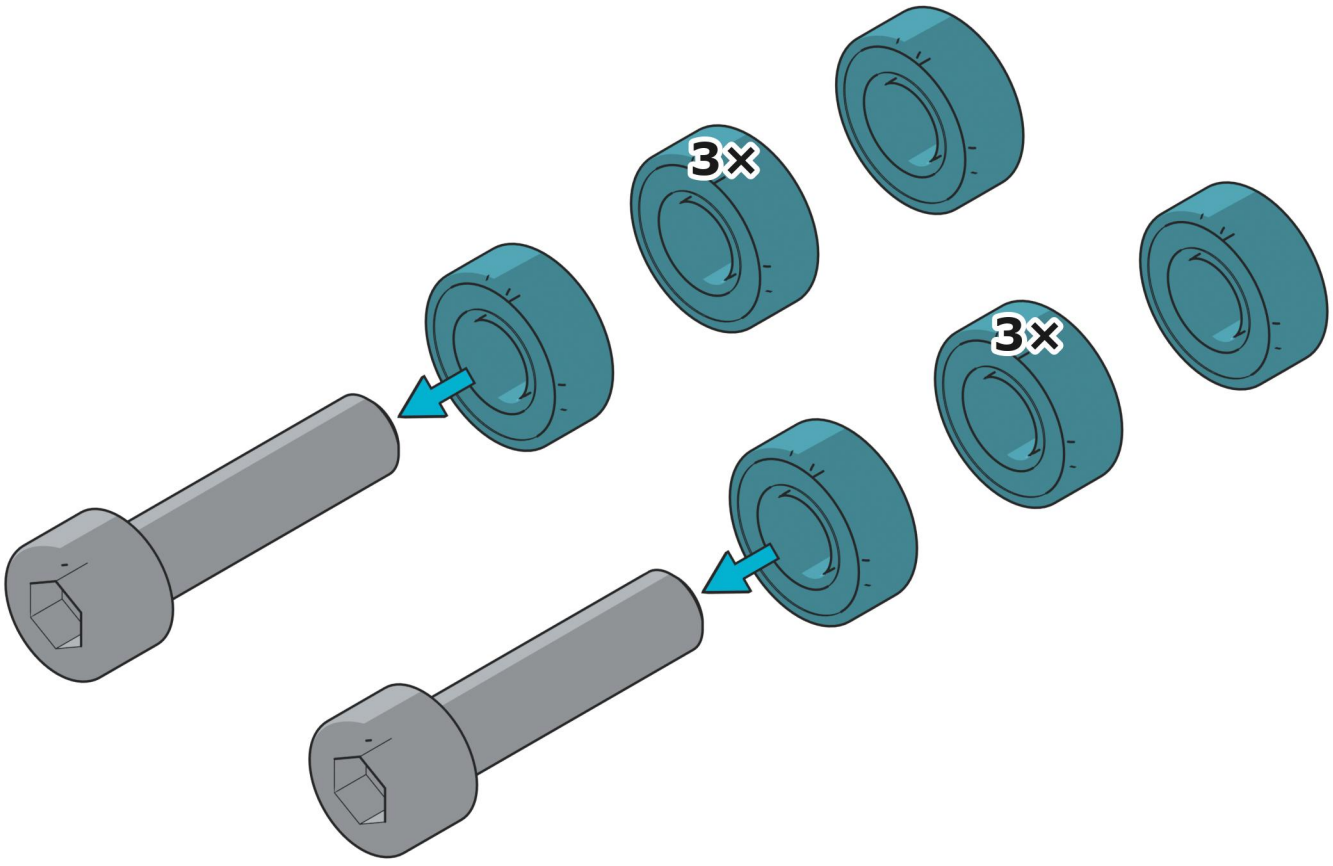


2



2×
M5×20 Screw

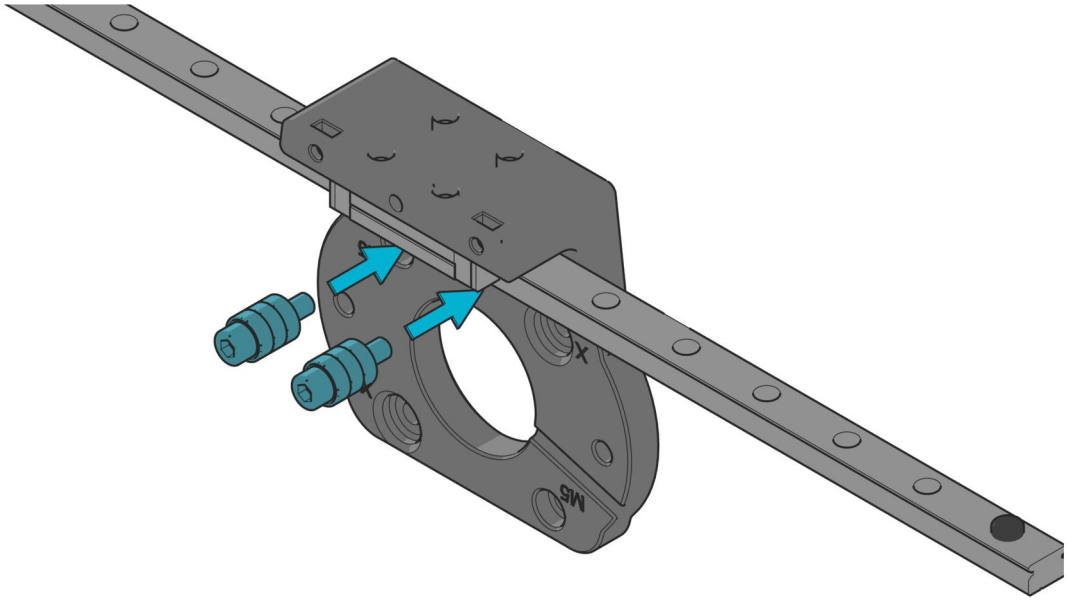
6×
5×11×4 Bearing

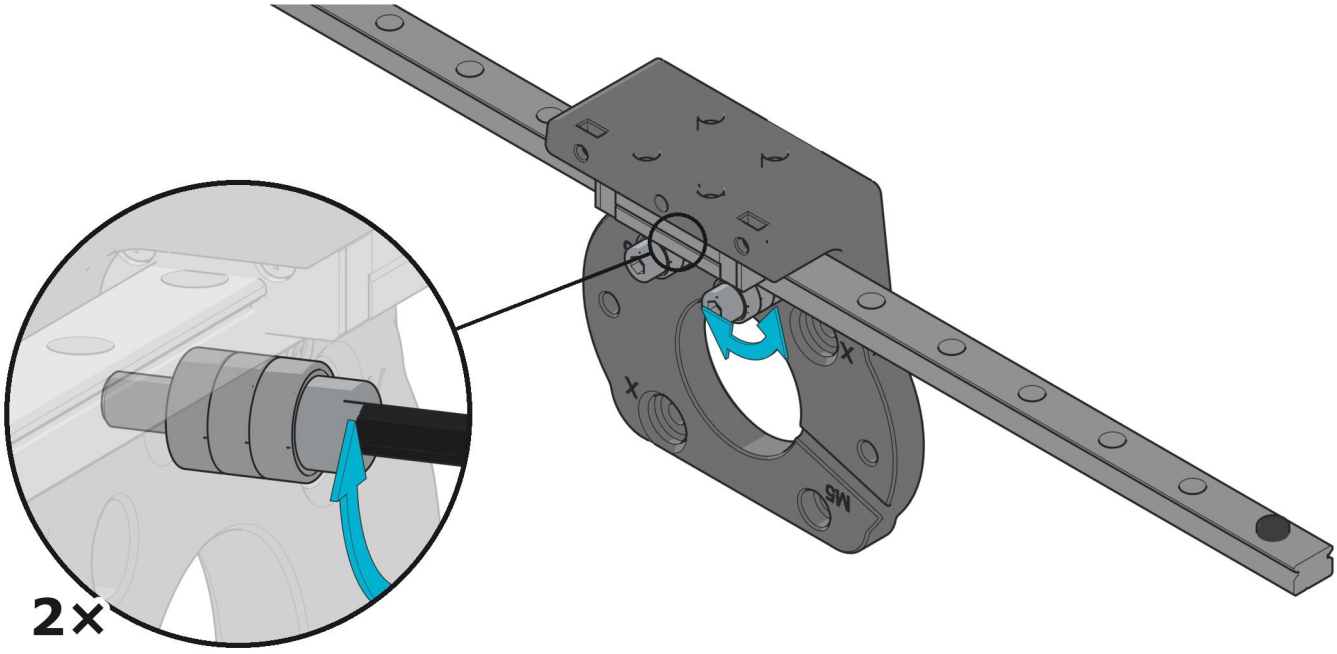


3



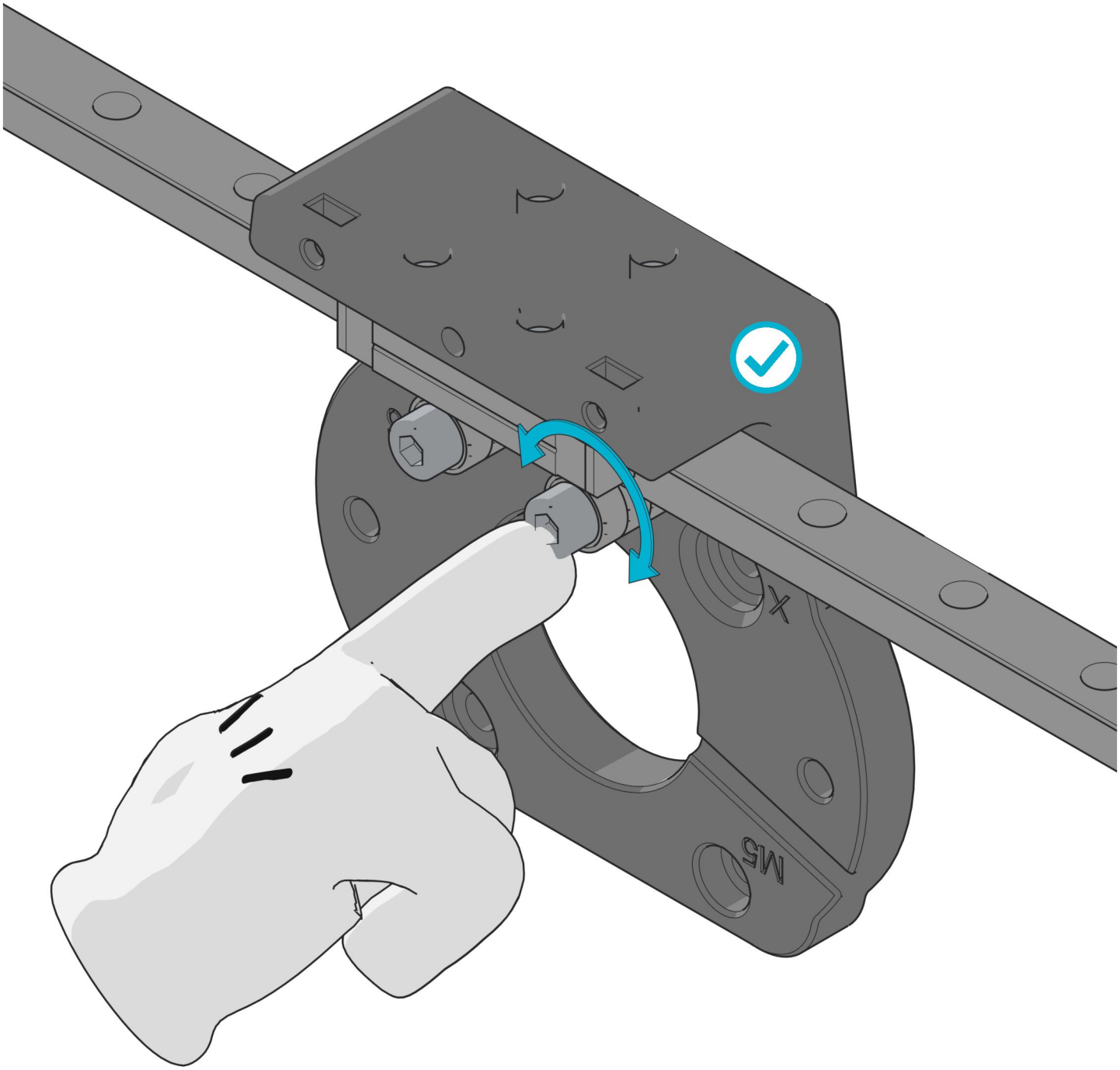
4 mm Hex key





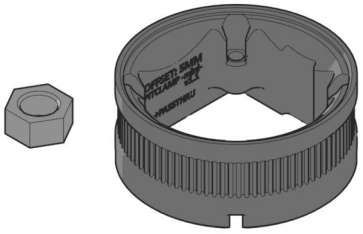
Snug only: do not over-tighten the idler screws.

4



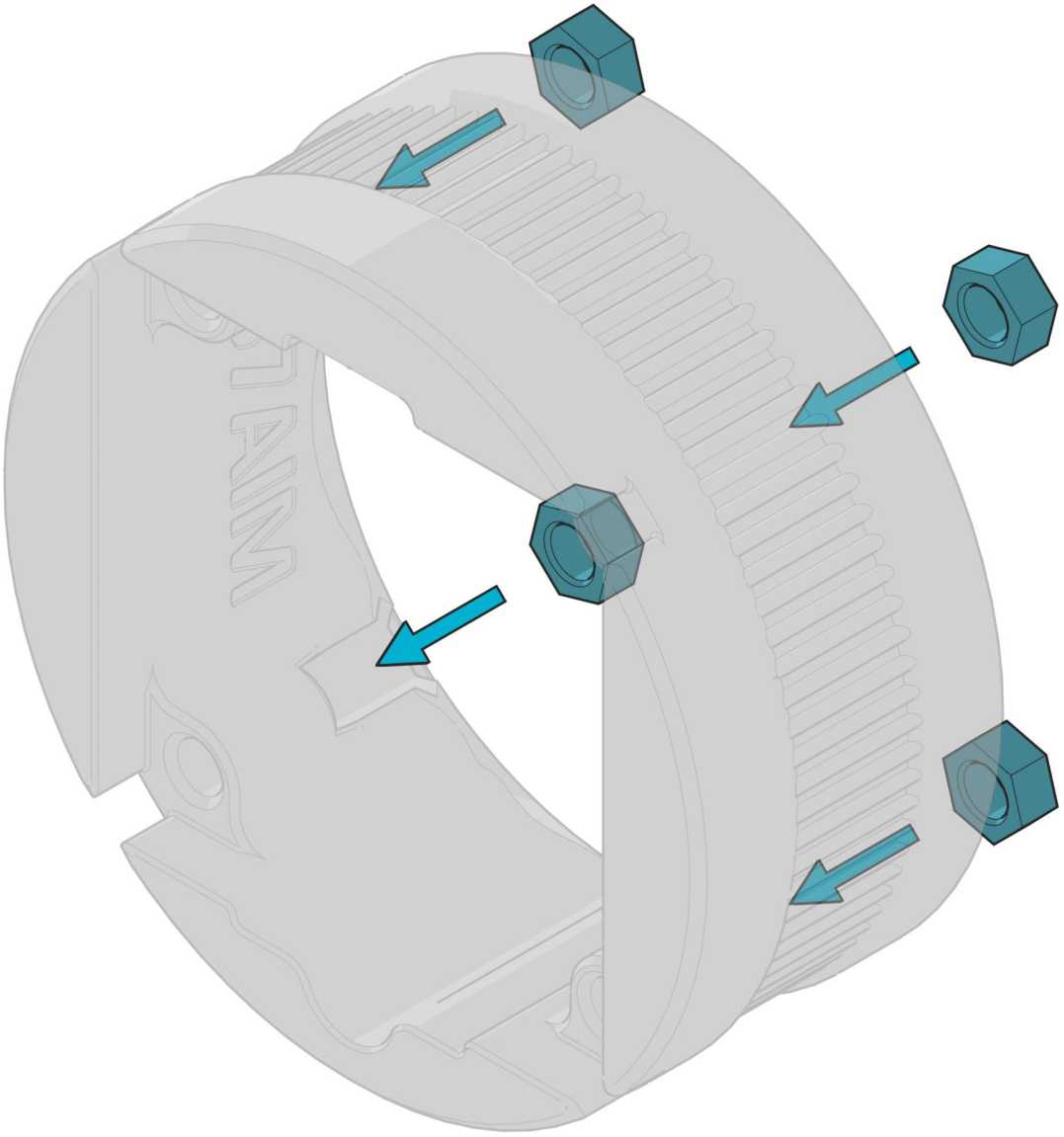
Snug only: each bearing must spin freely. If one does not, loosen its screw a little.

5



4× M5 Nut

1× Motor ring

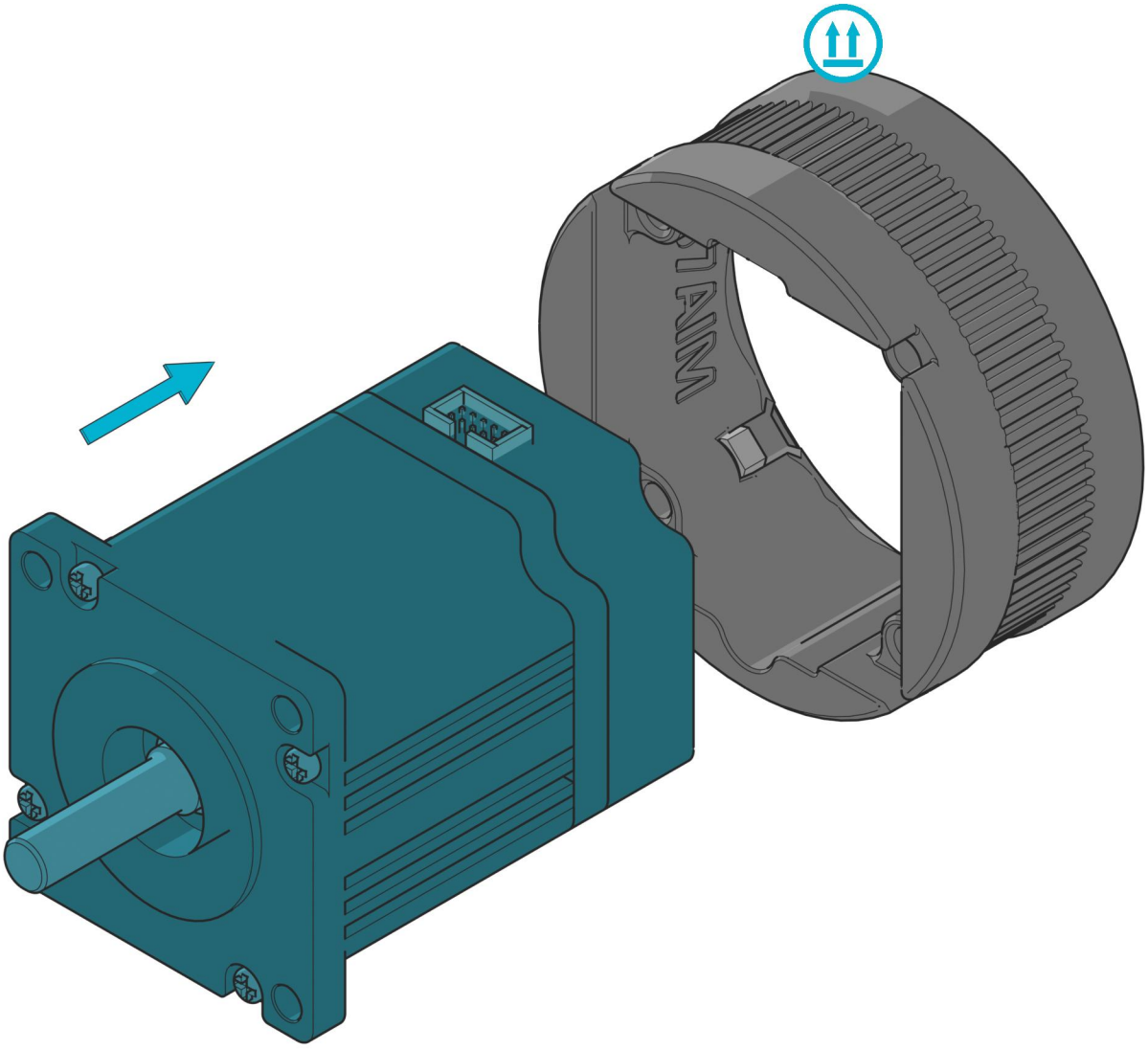


6

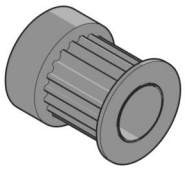


1×

Servo motor



7



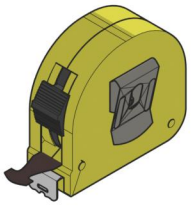
10 mm

1×

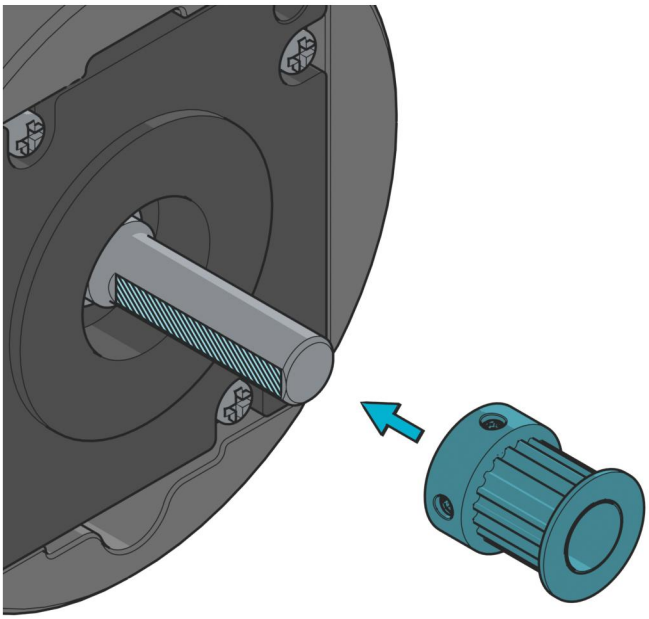
Pulley

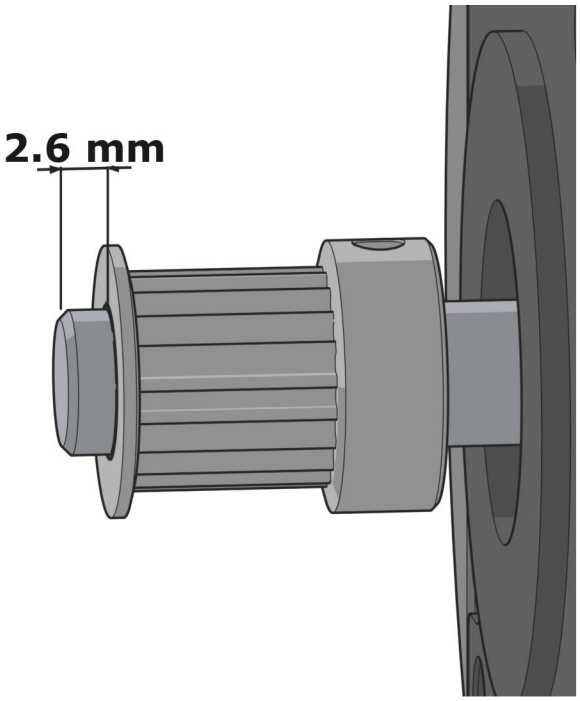


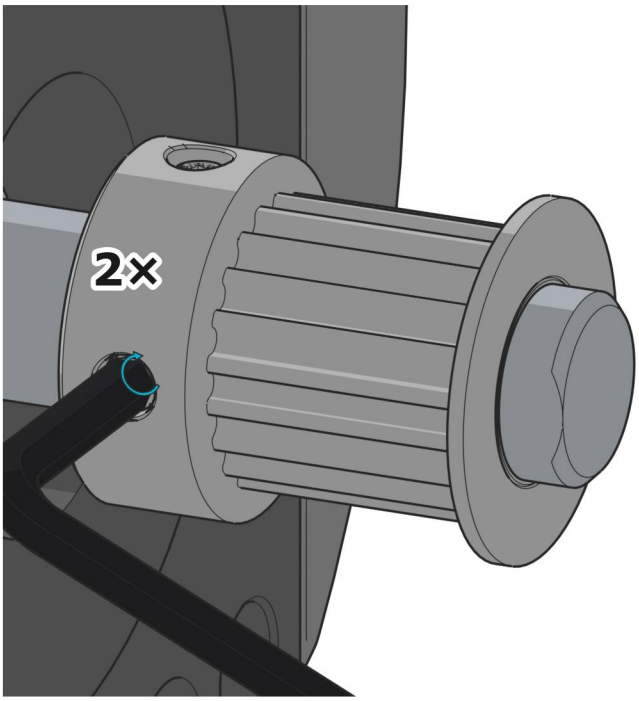
2 mm Hex key

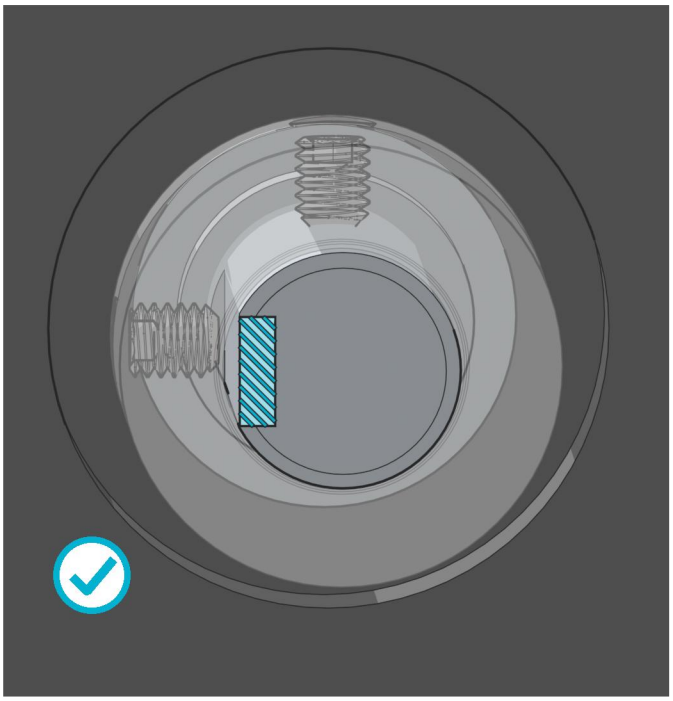


Tape measure

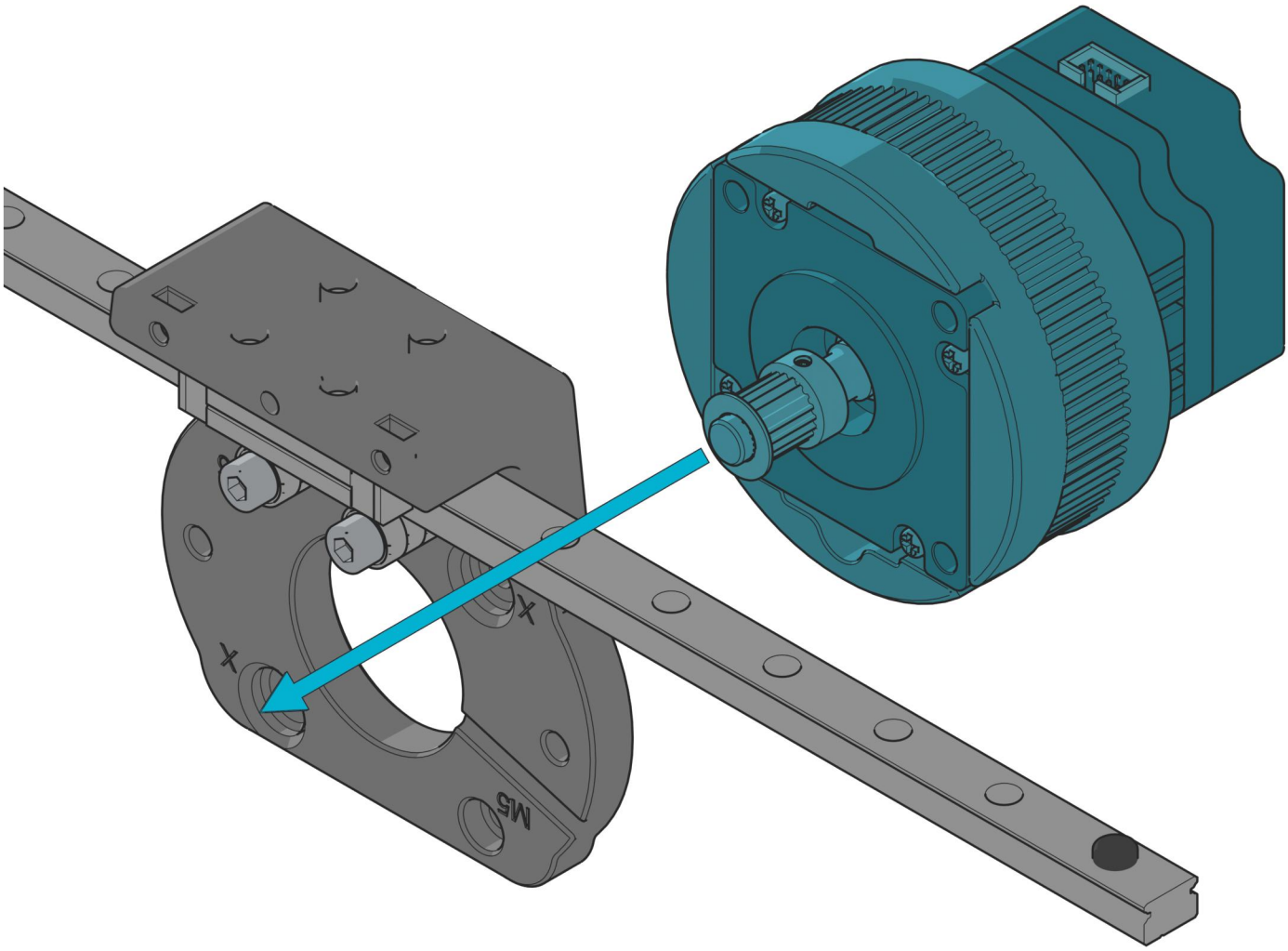




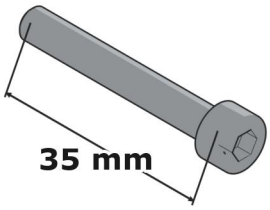




8



9

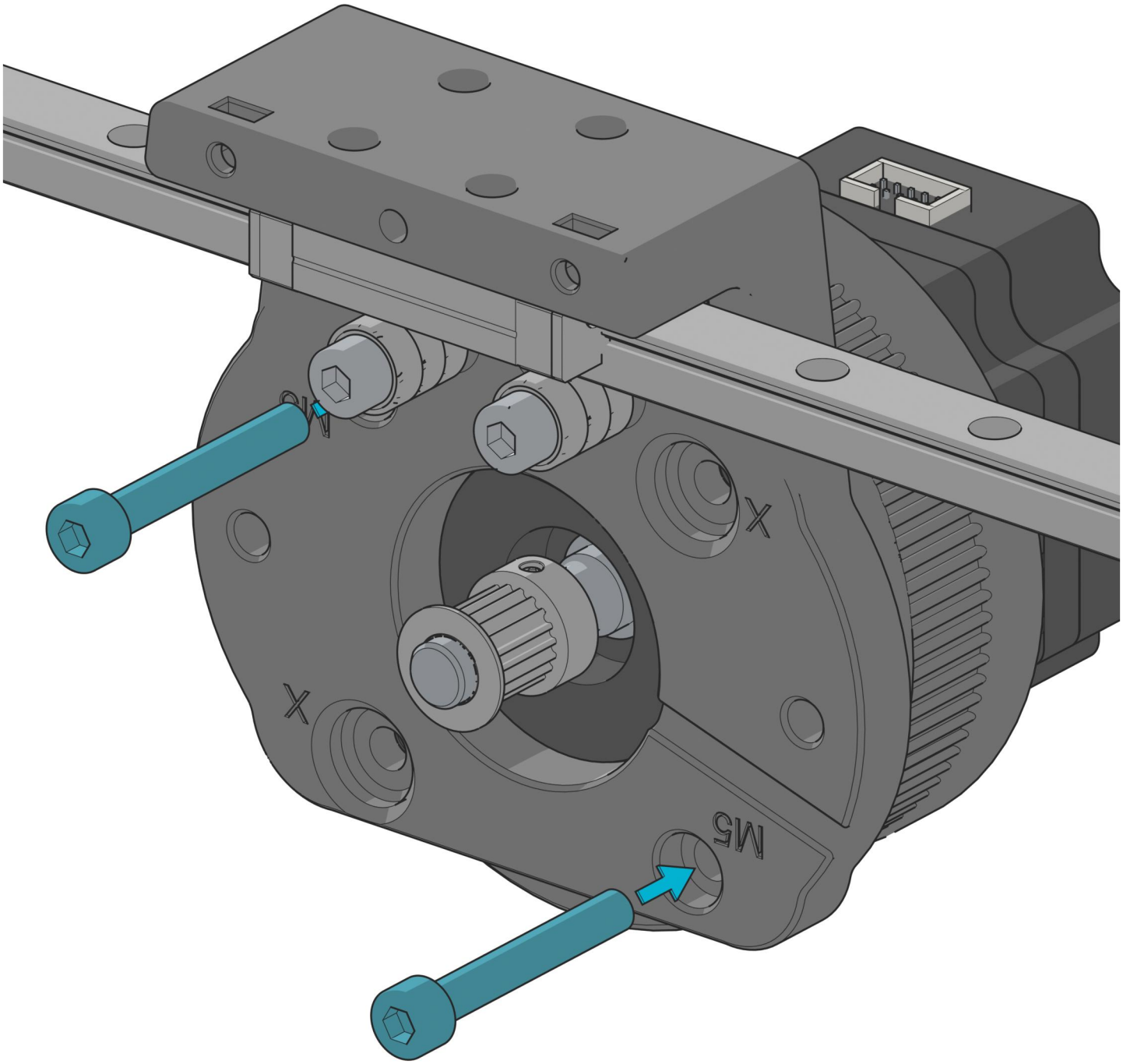


2×

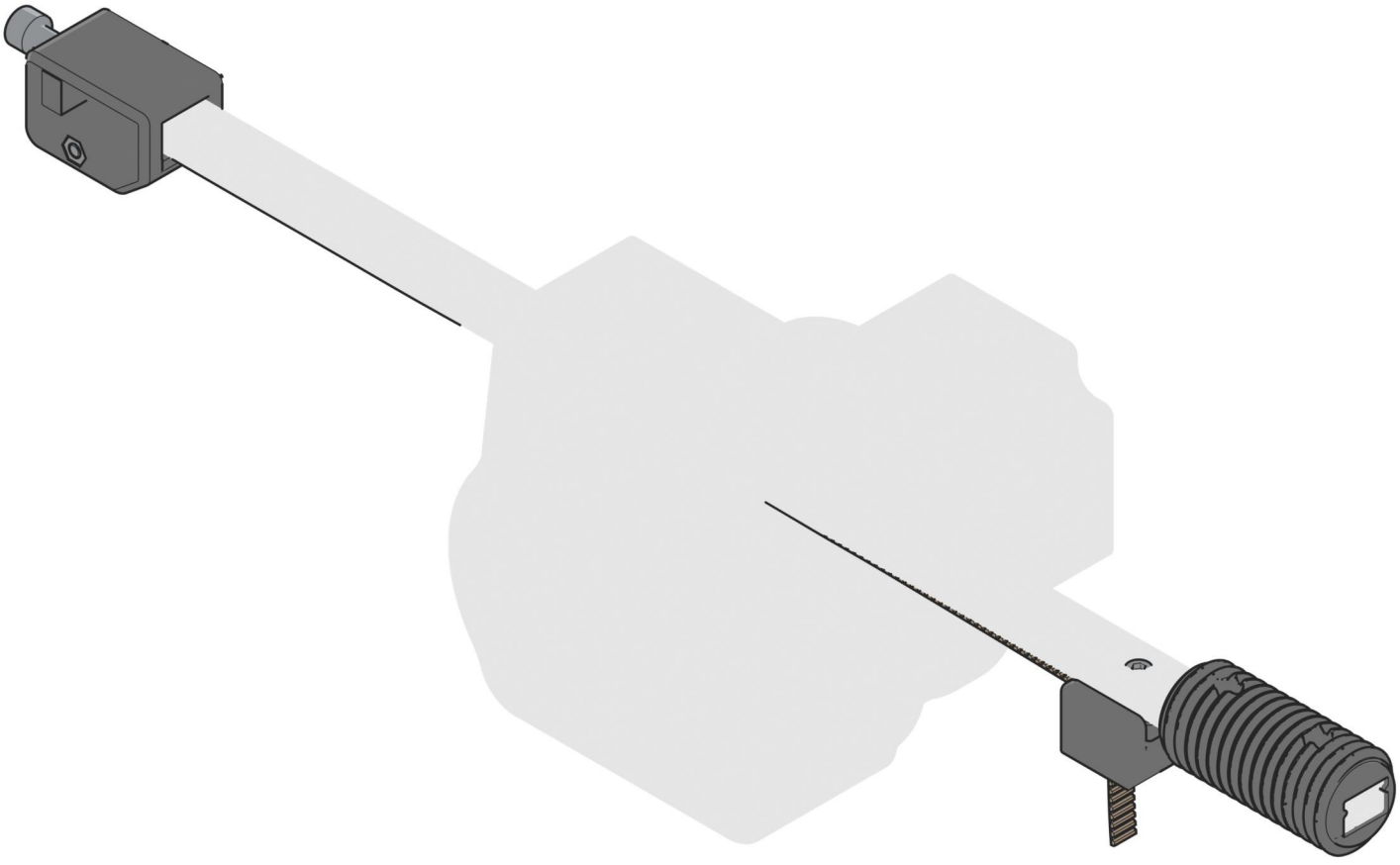
M5×35 Screw



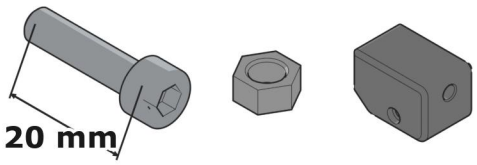
4 mm Hex key



2 The belt



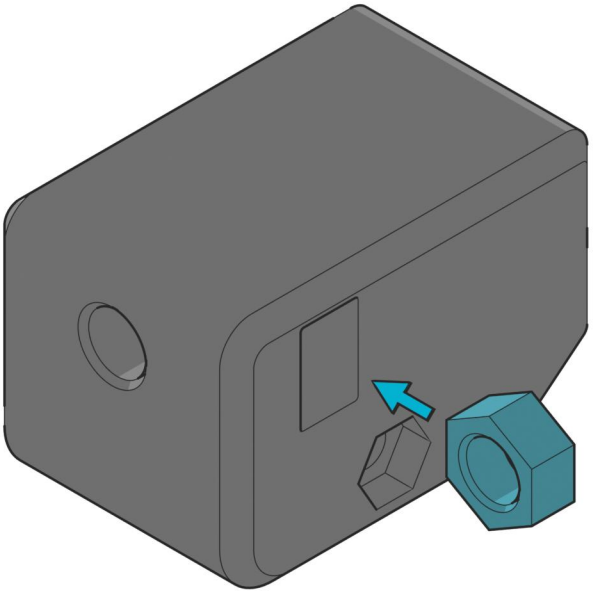
10

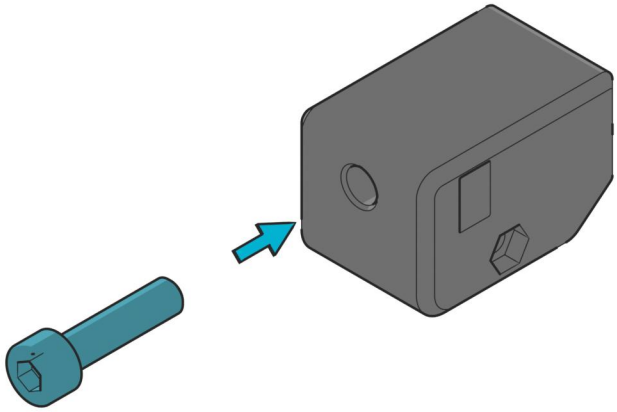


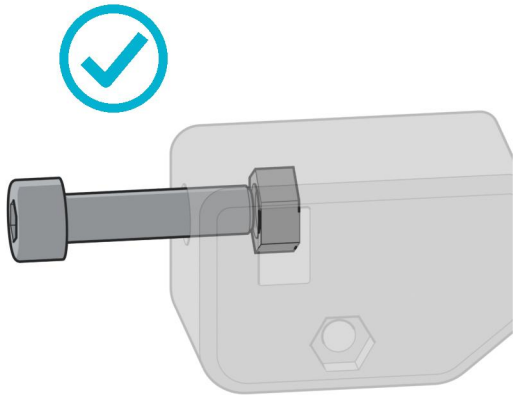
1× M5×20 Screw

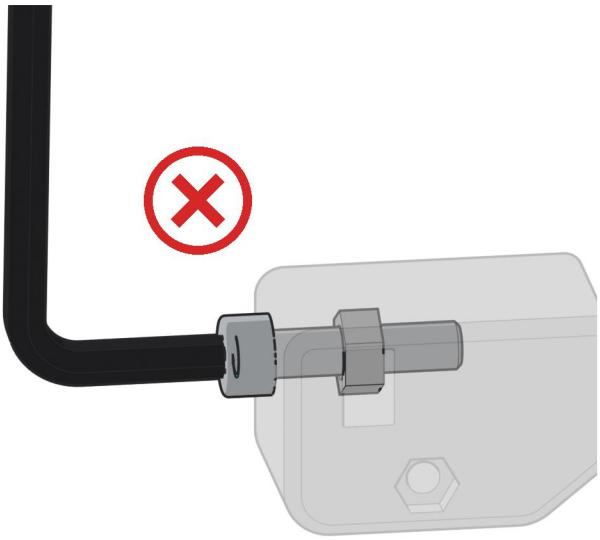
1× M5 Nut

1× Belt tensioner

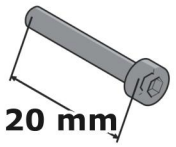








11

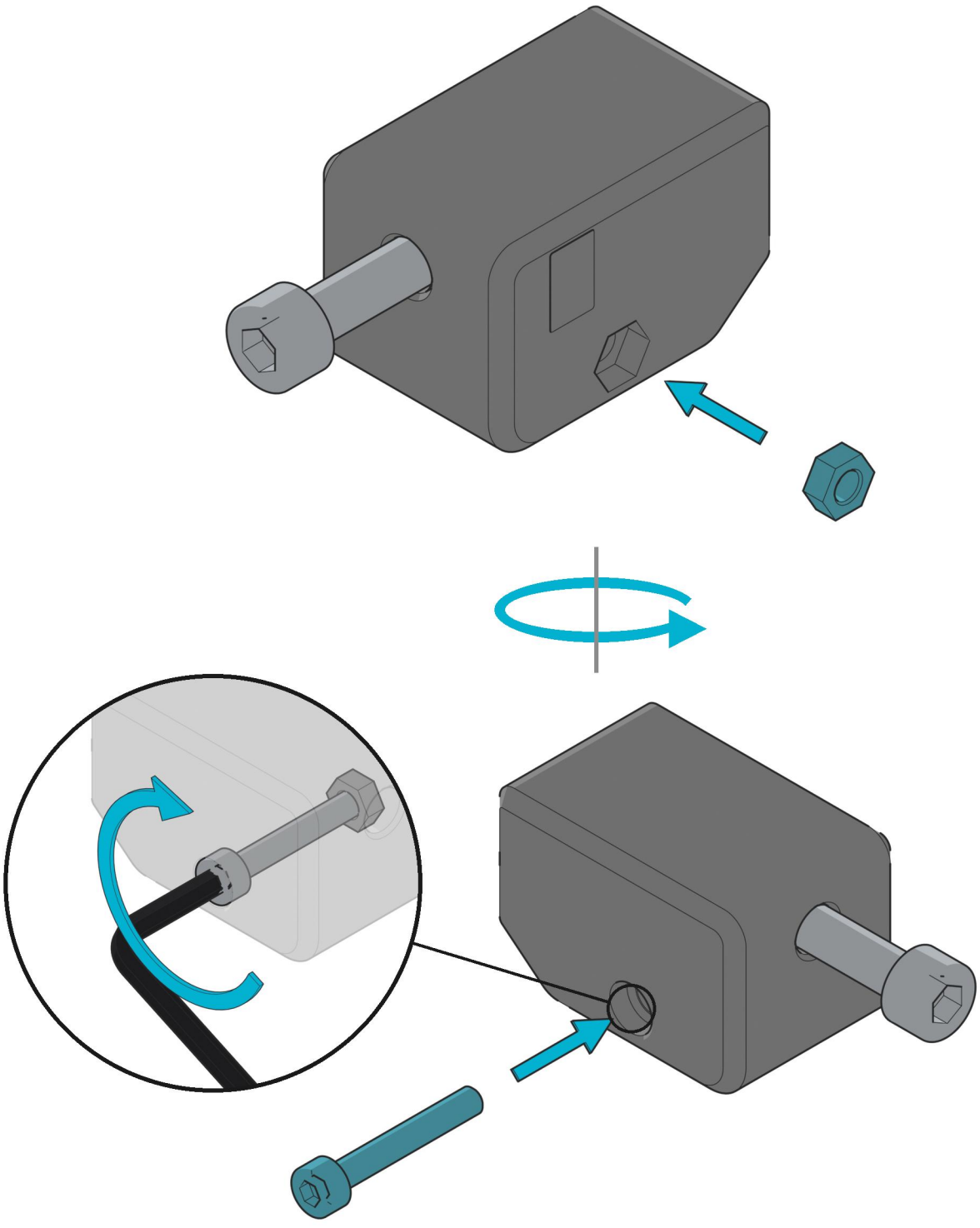


1× M3×20 Screw

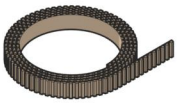
1× M3 Nut



2.5 mm Hex key



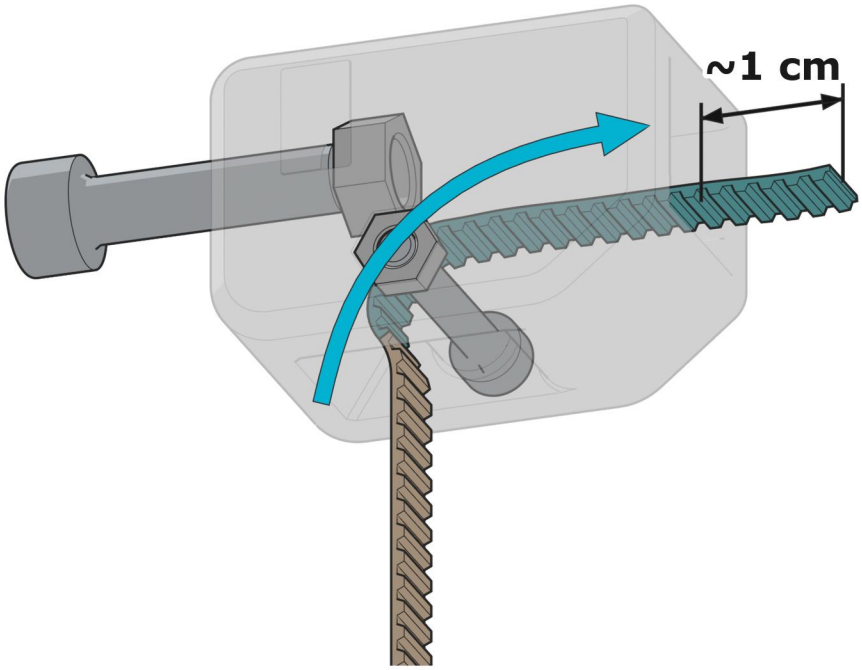
12

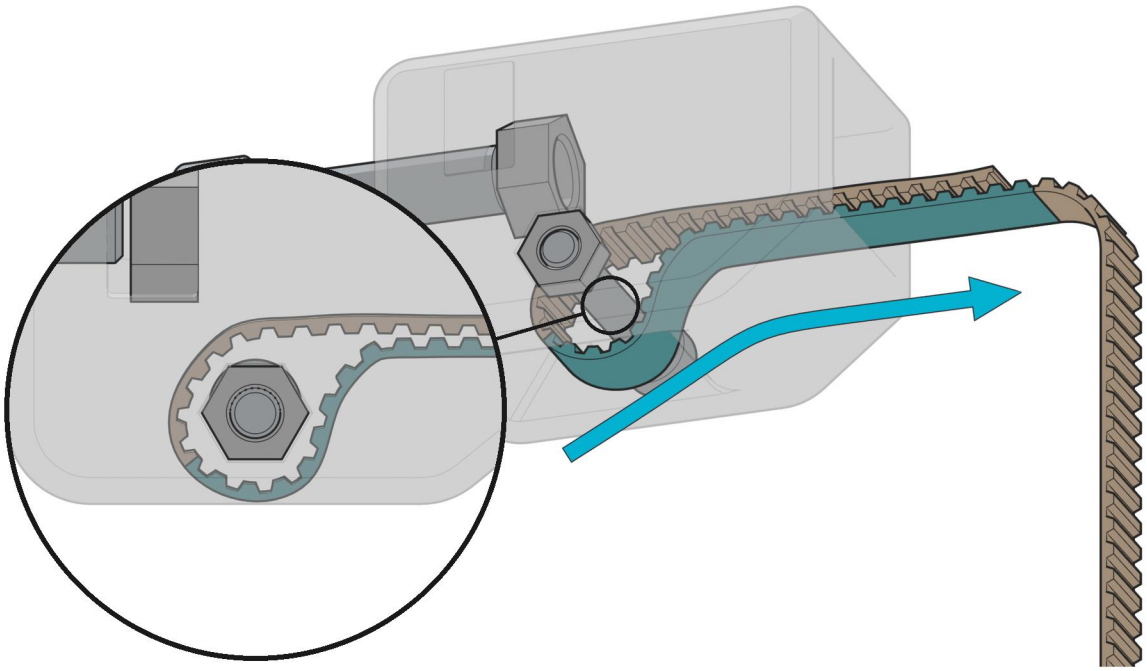


500 mm

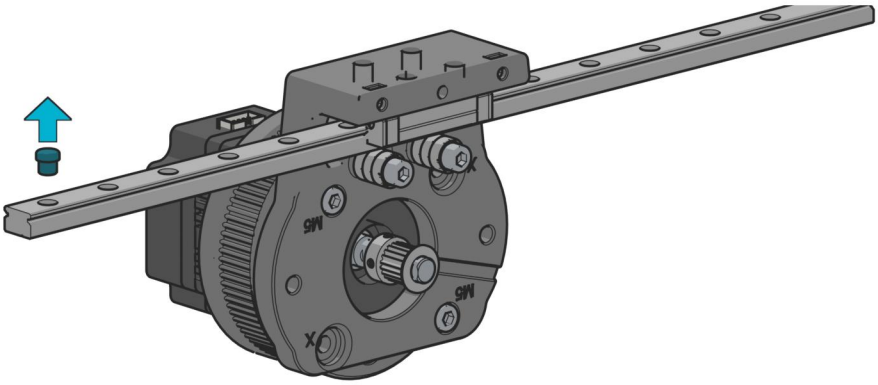
1×

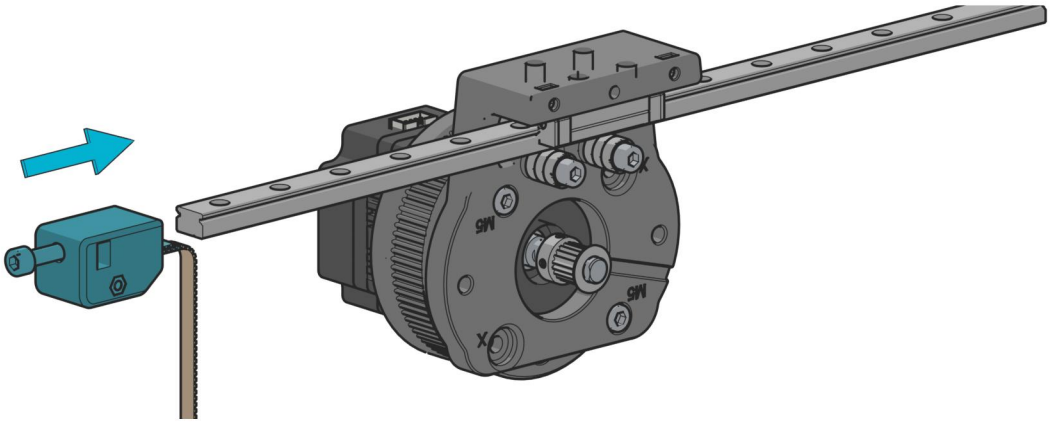
Belt

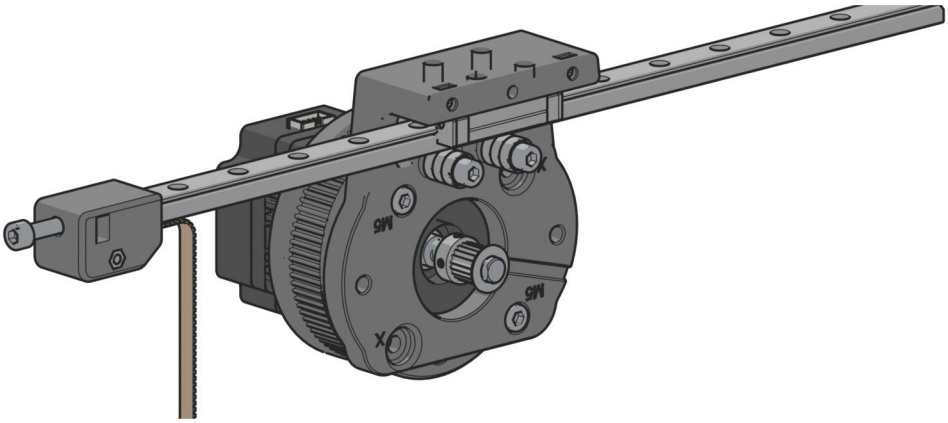




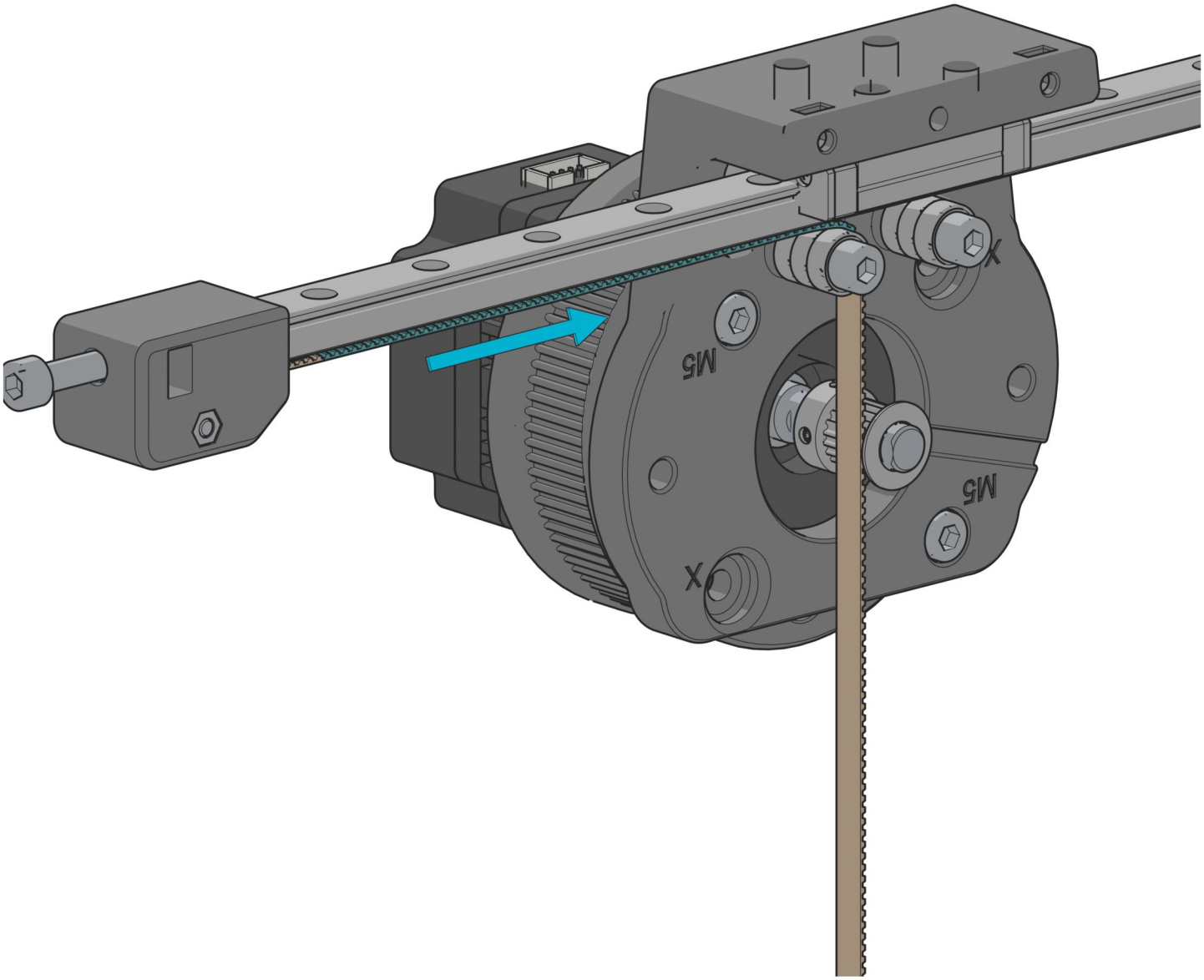
13



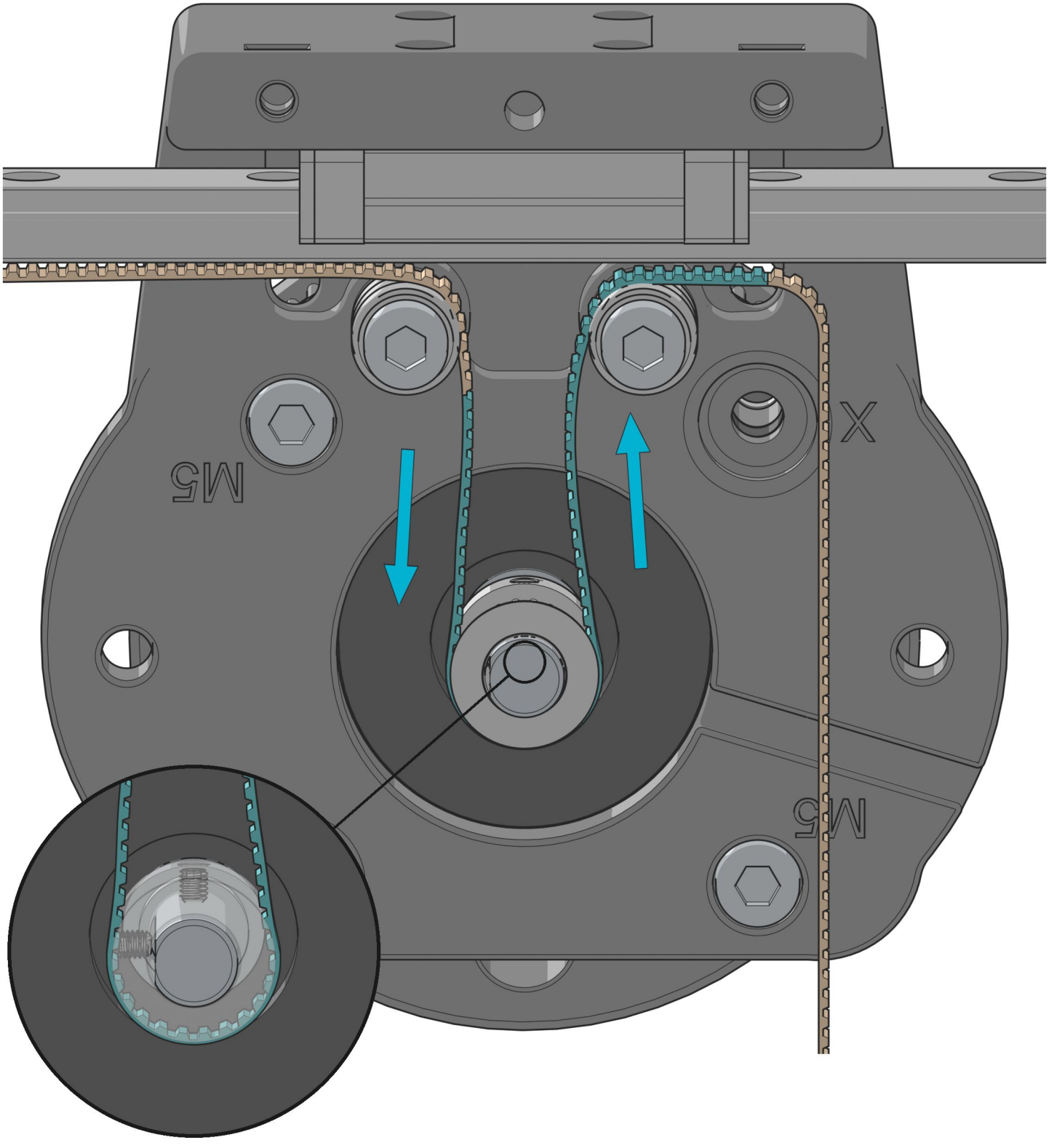




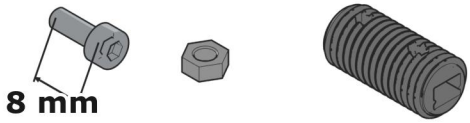
14



15



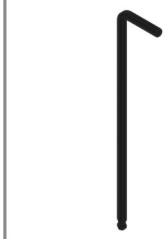
16



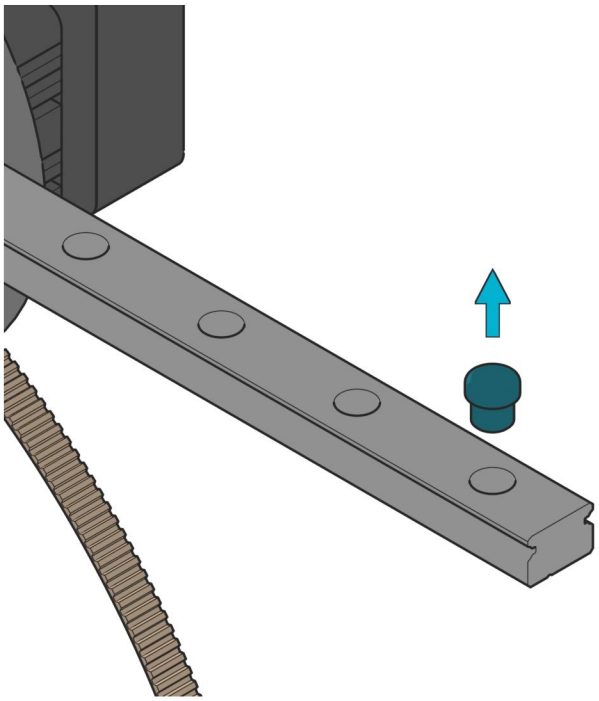
2× M3×8 Screw

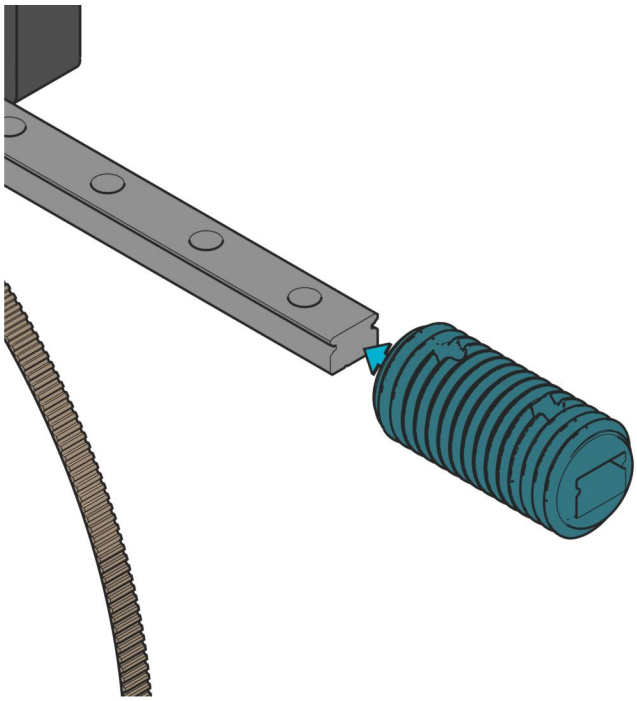
2× M3 Nut

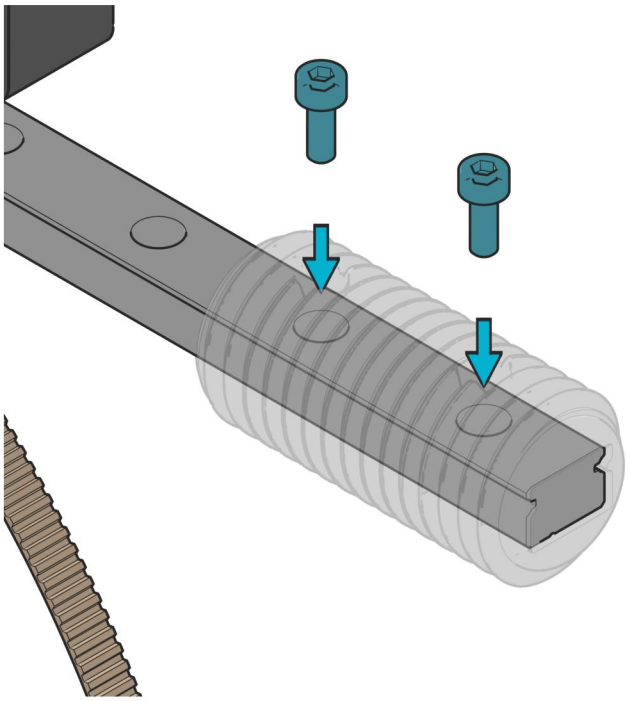
1× 24 mm Threaded end effector

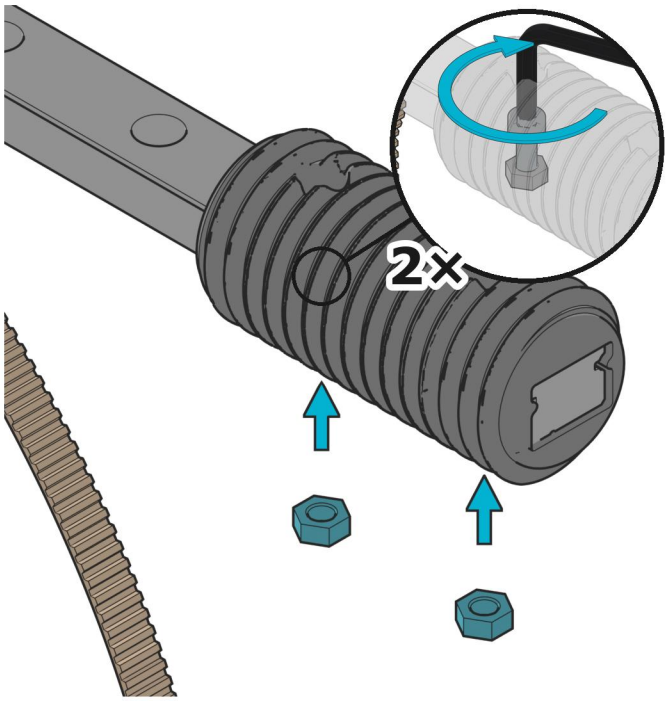


2.5 mm Hex key

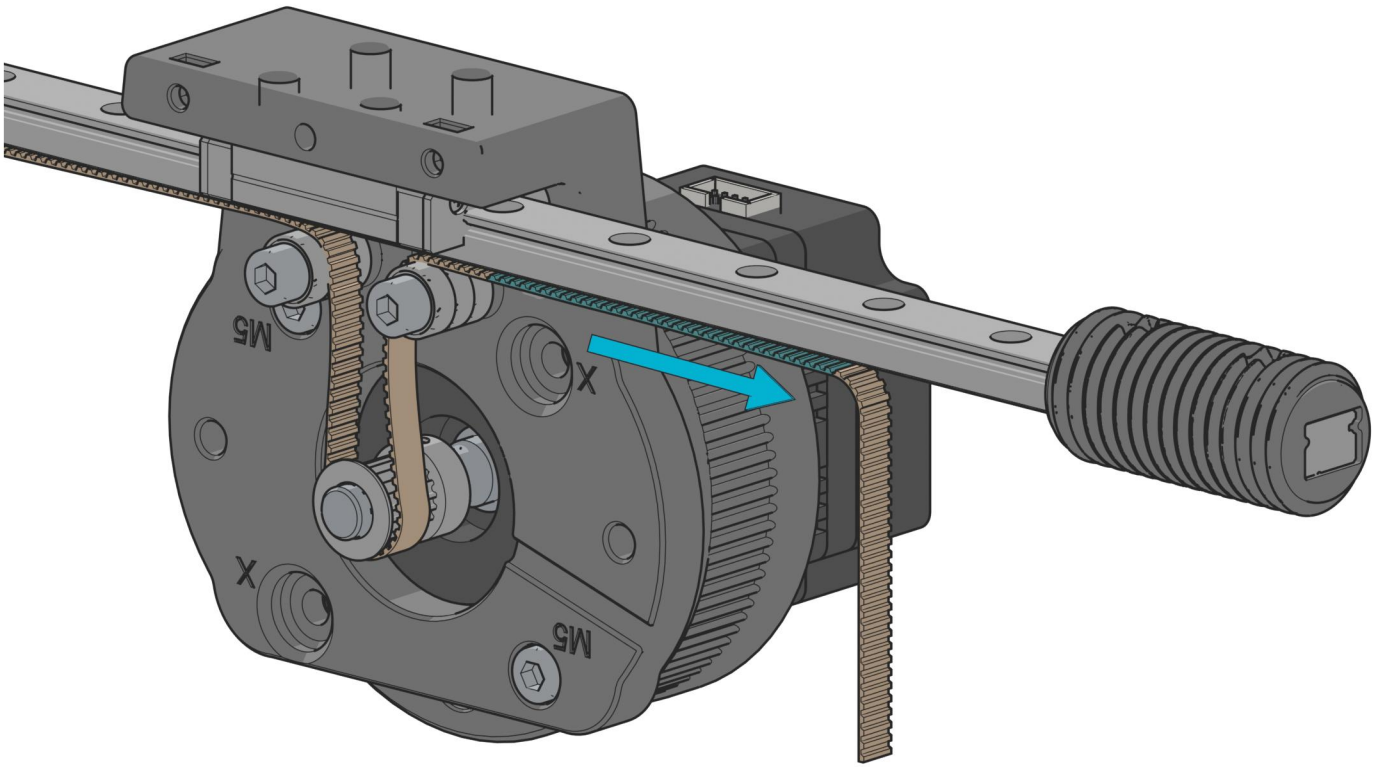








17

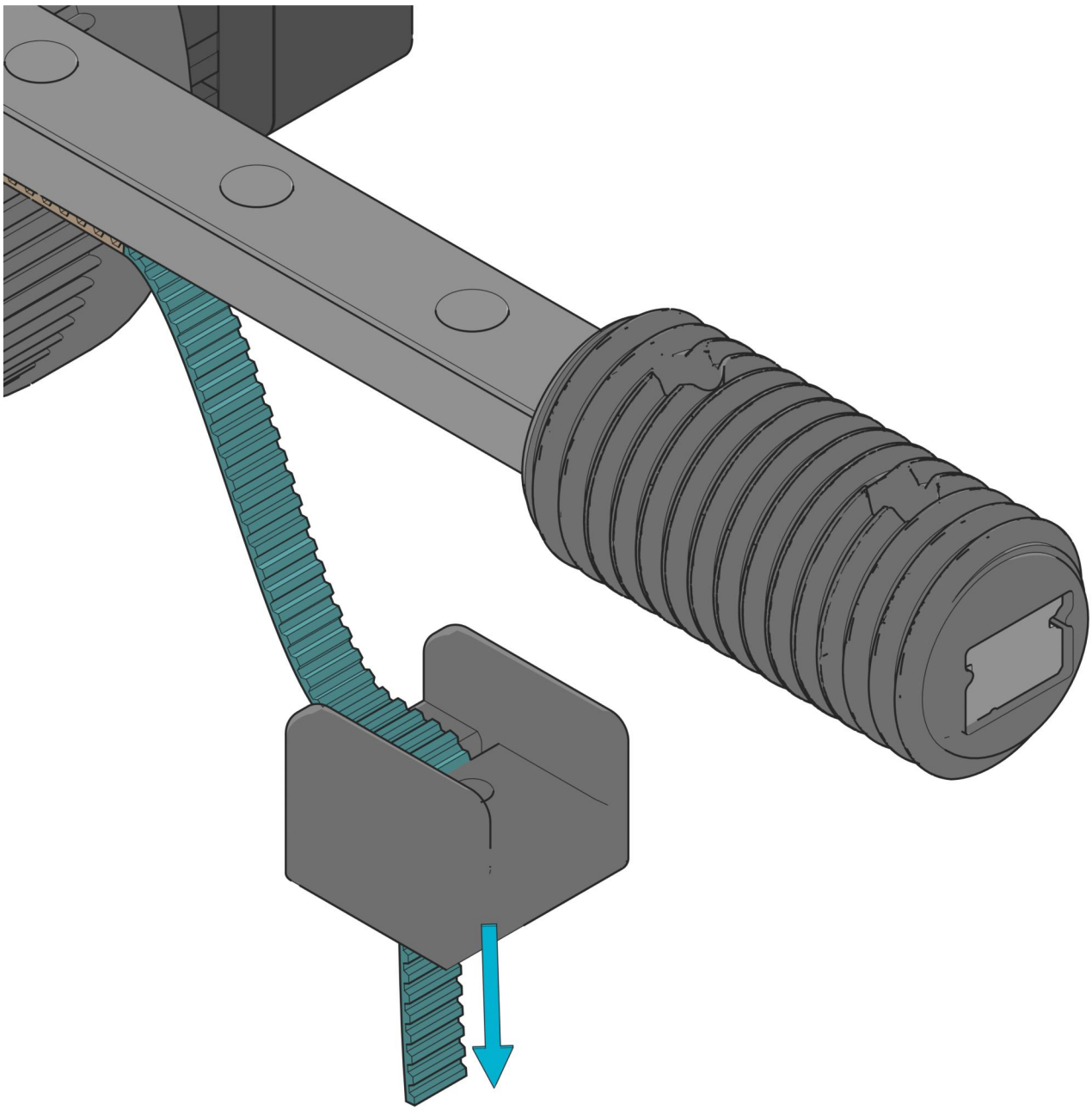


18



1×

Belt clamp
bottom

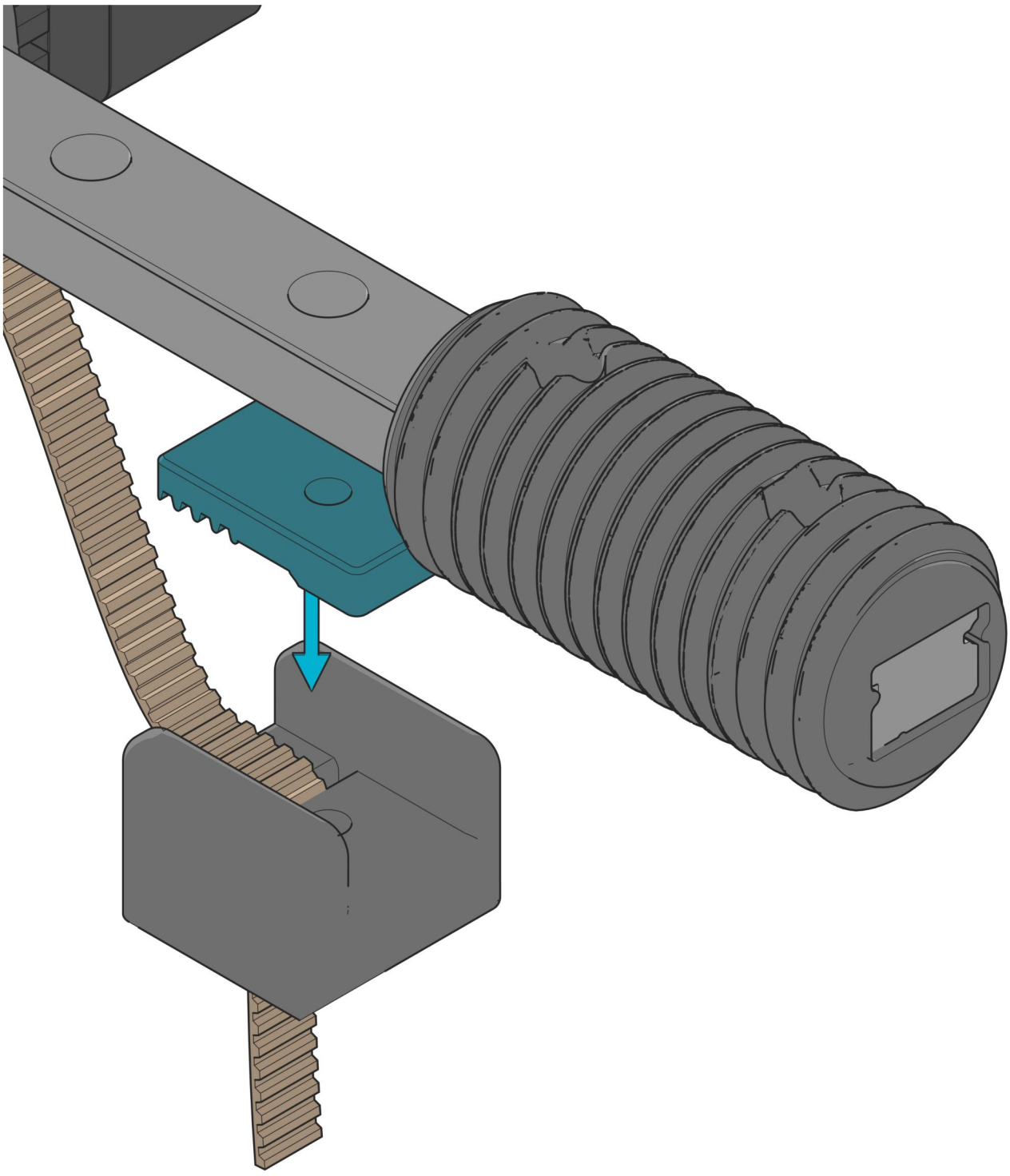


19

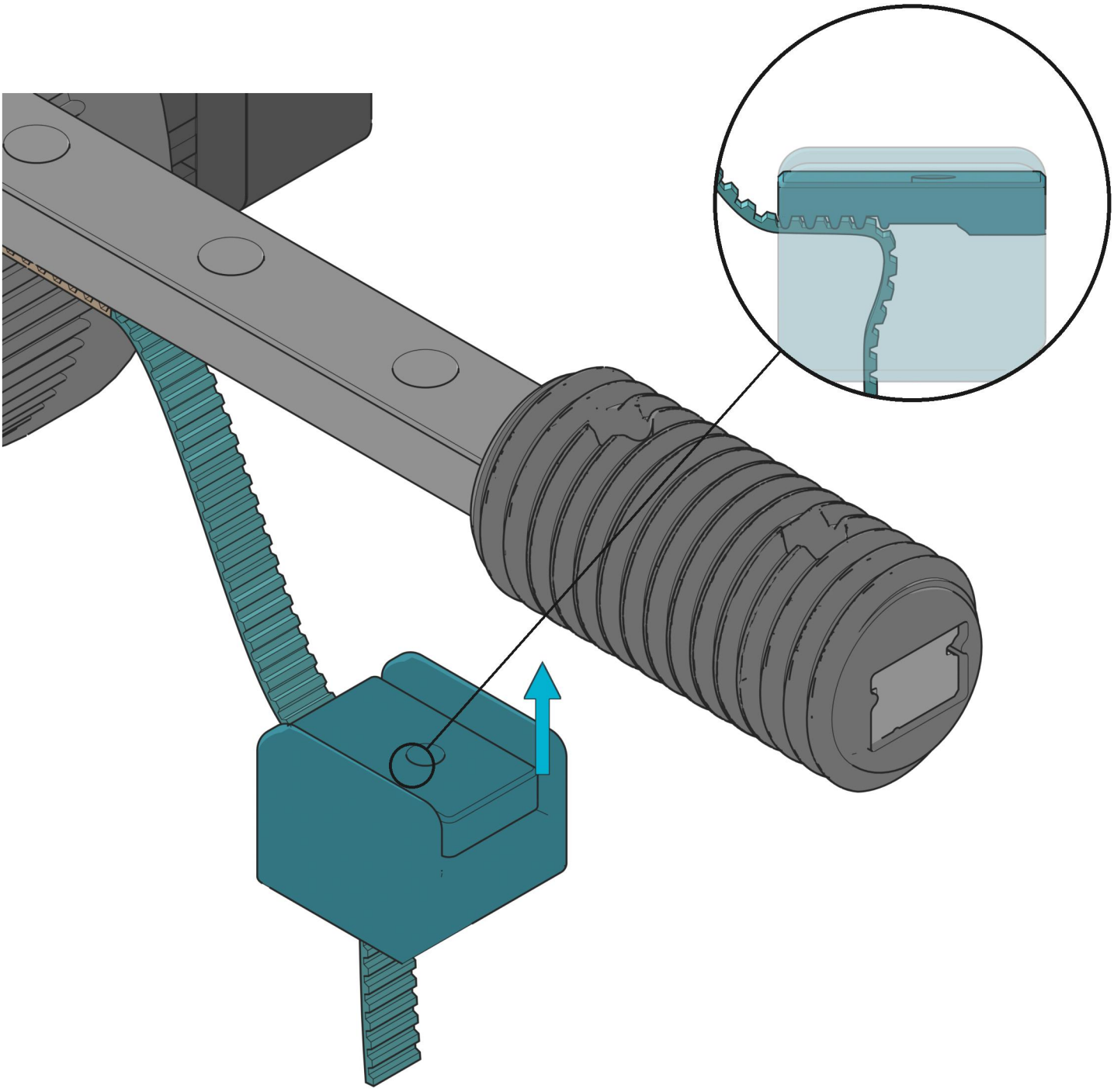


1×

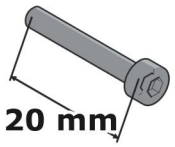
Belt clamp top



20



21

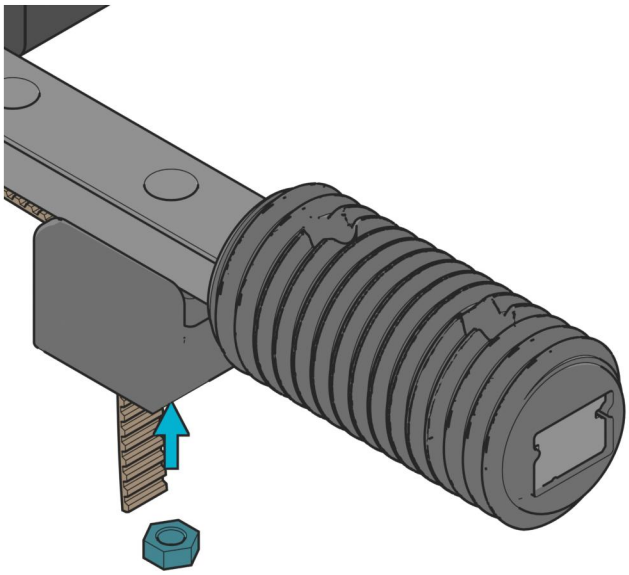


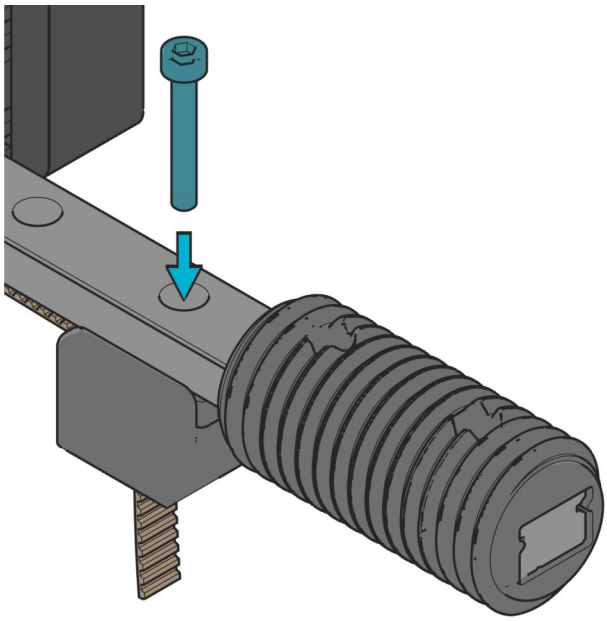
1× M3×20 Screw

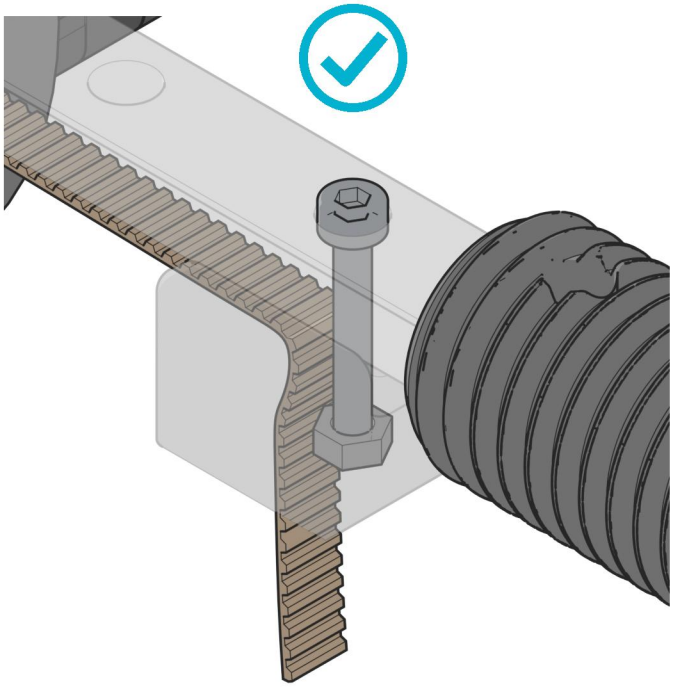
1× M3 Nut

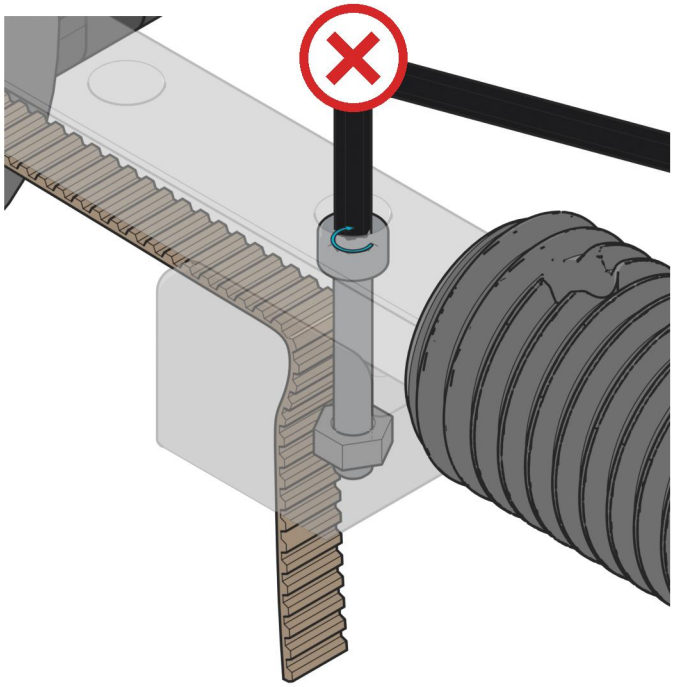


2.5 mm Hex key

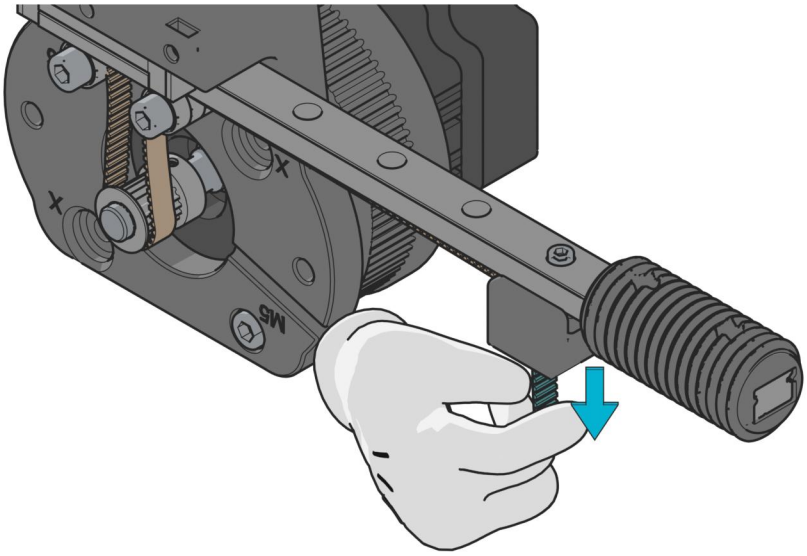


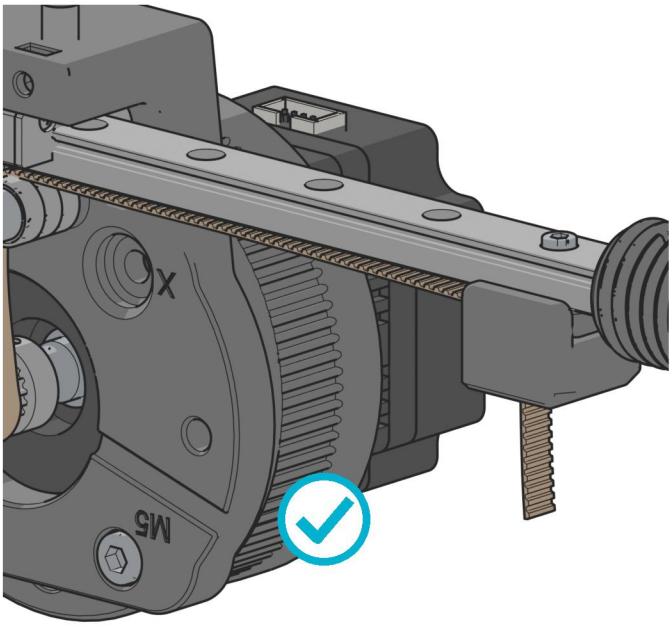


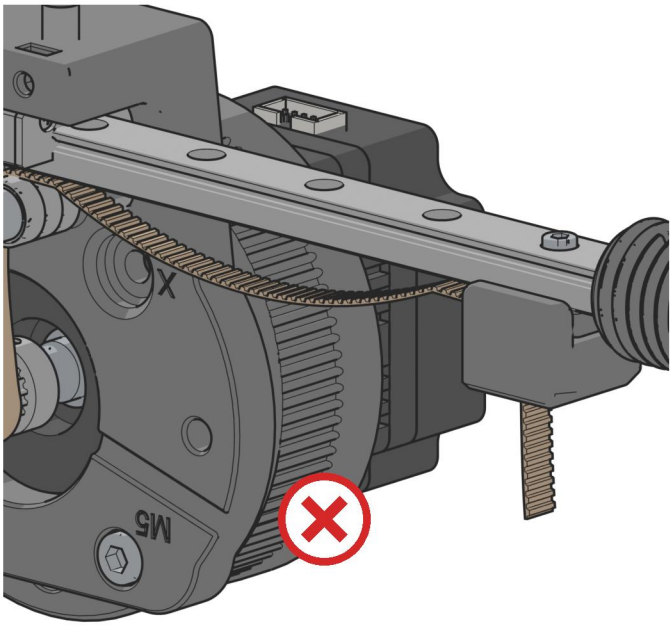




22





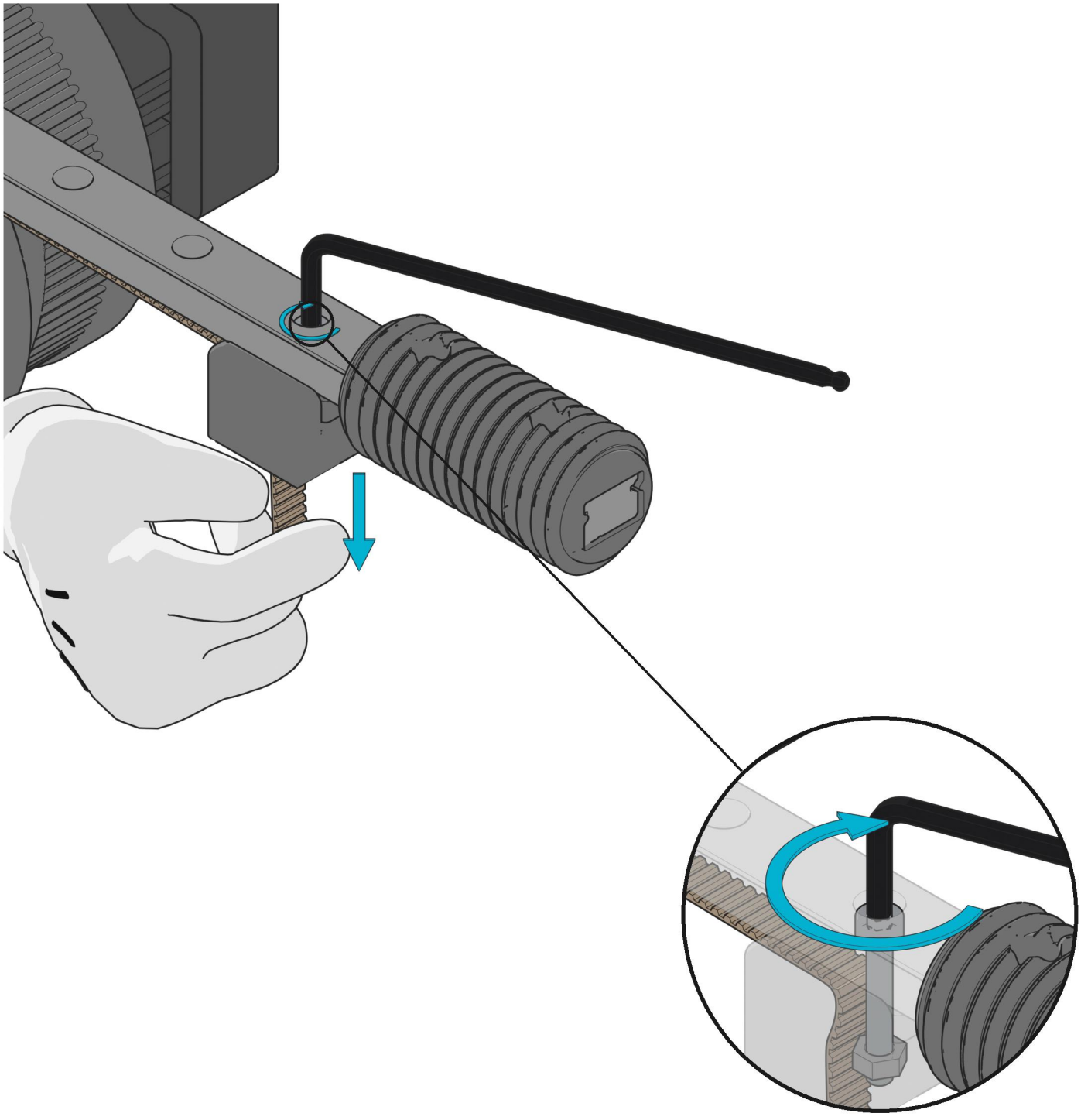


Pull the belt's end down until the belt is taut.

23

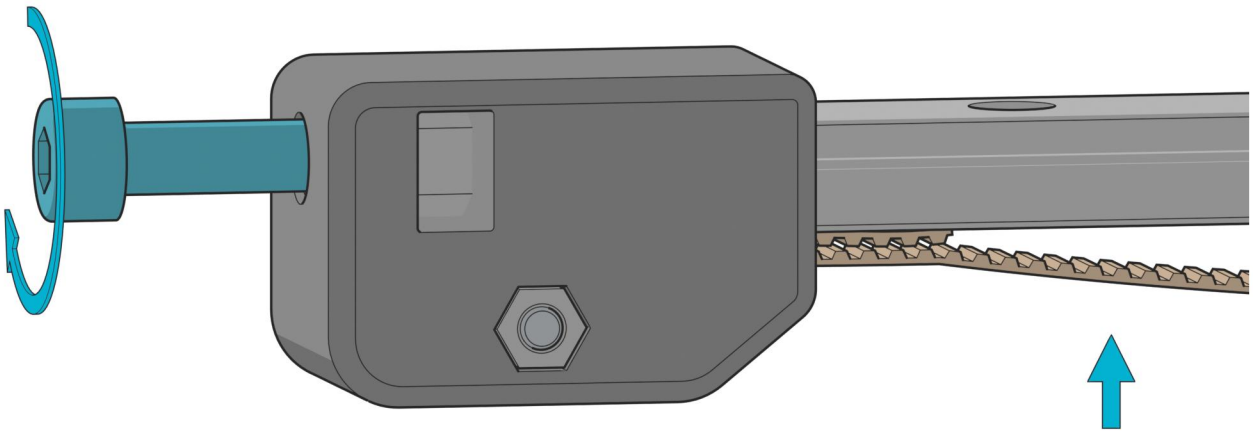


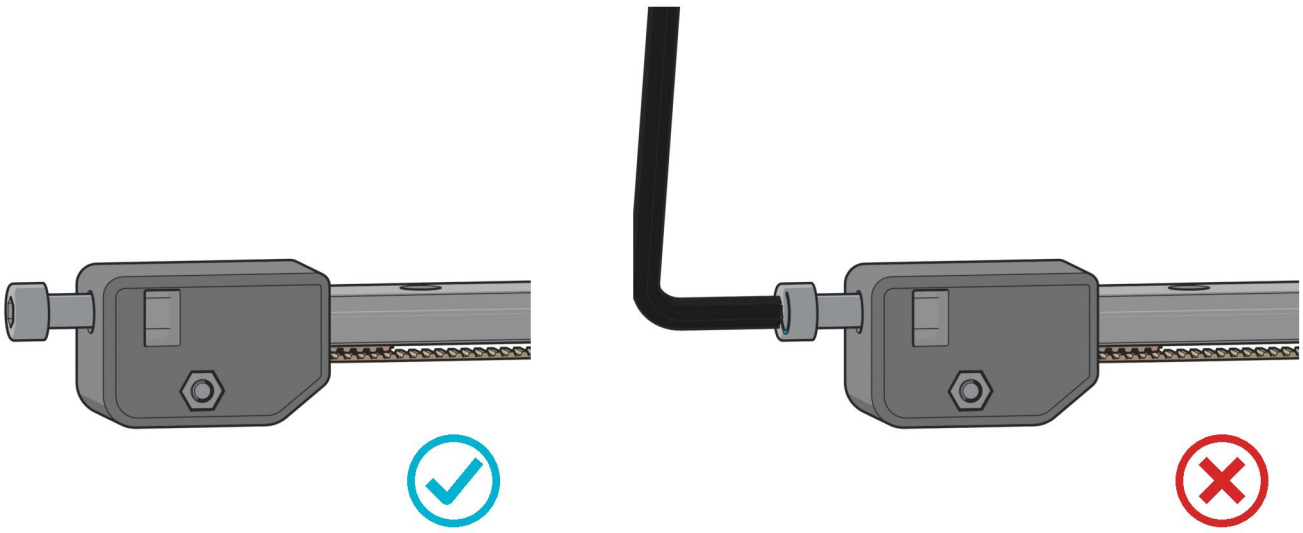
2.5 mm Hex key



Keep pulling the belt taut while you tighten.

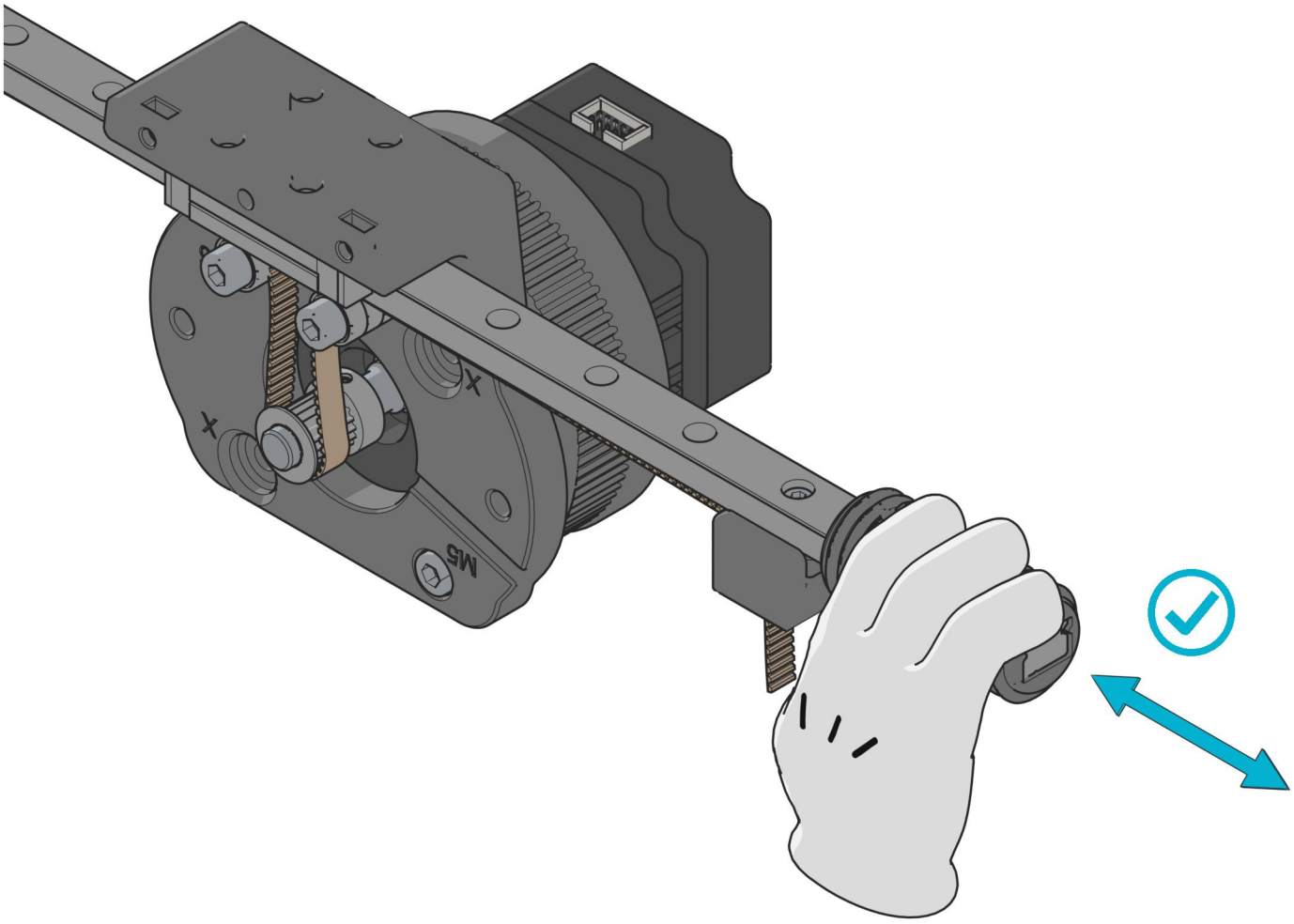
24





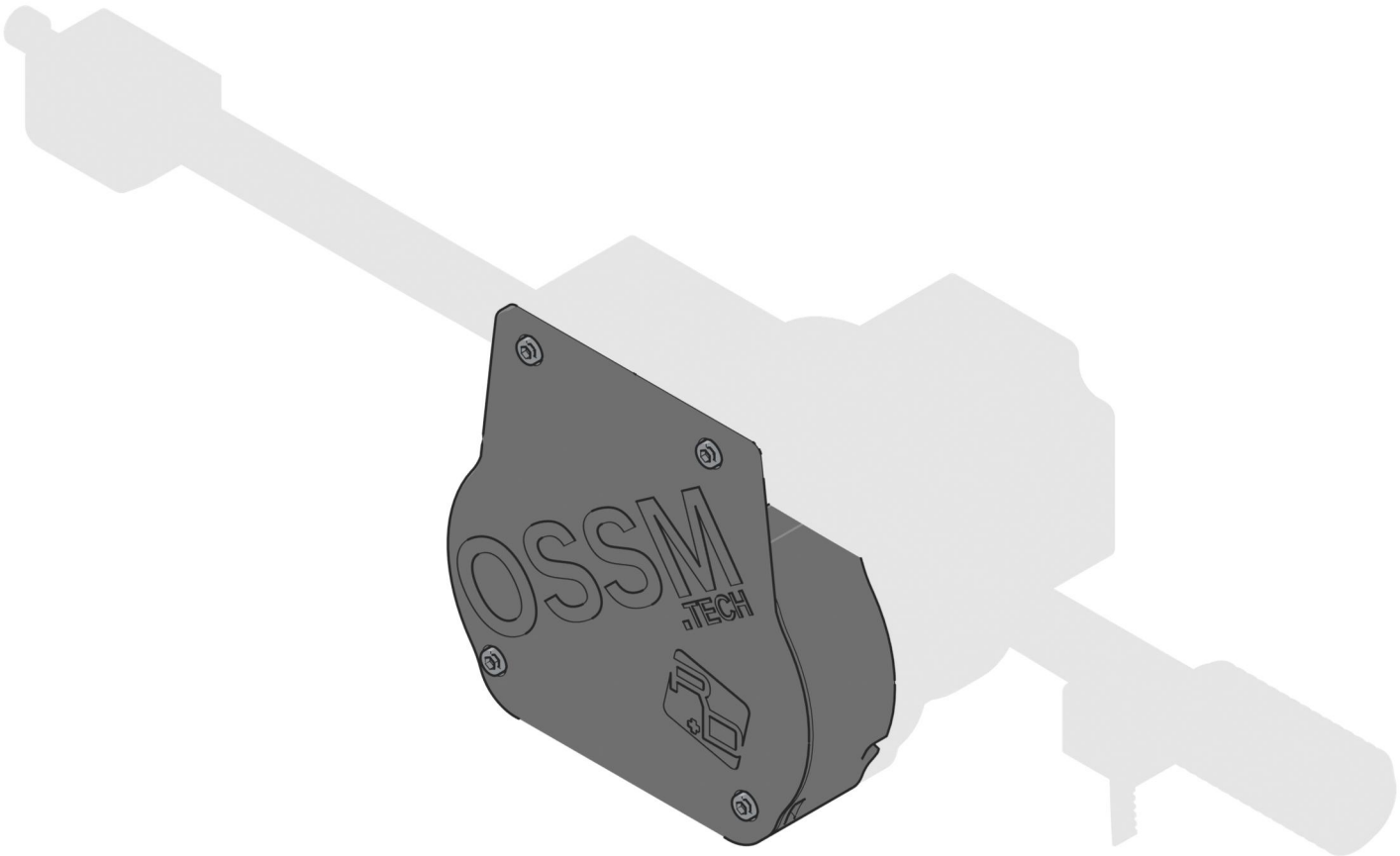
Turn it in by hand: do not over-tighten.

25

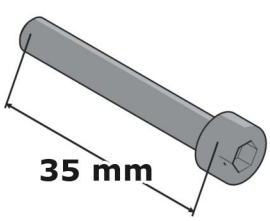


Push and pull the end effector: the rail must slide smoothly through the head.

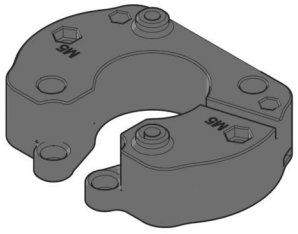
3 Closing the head



26



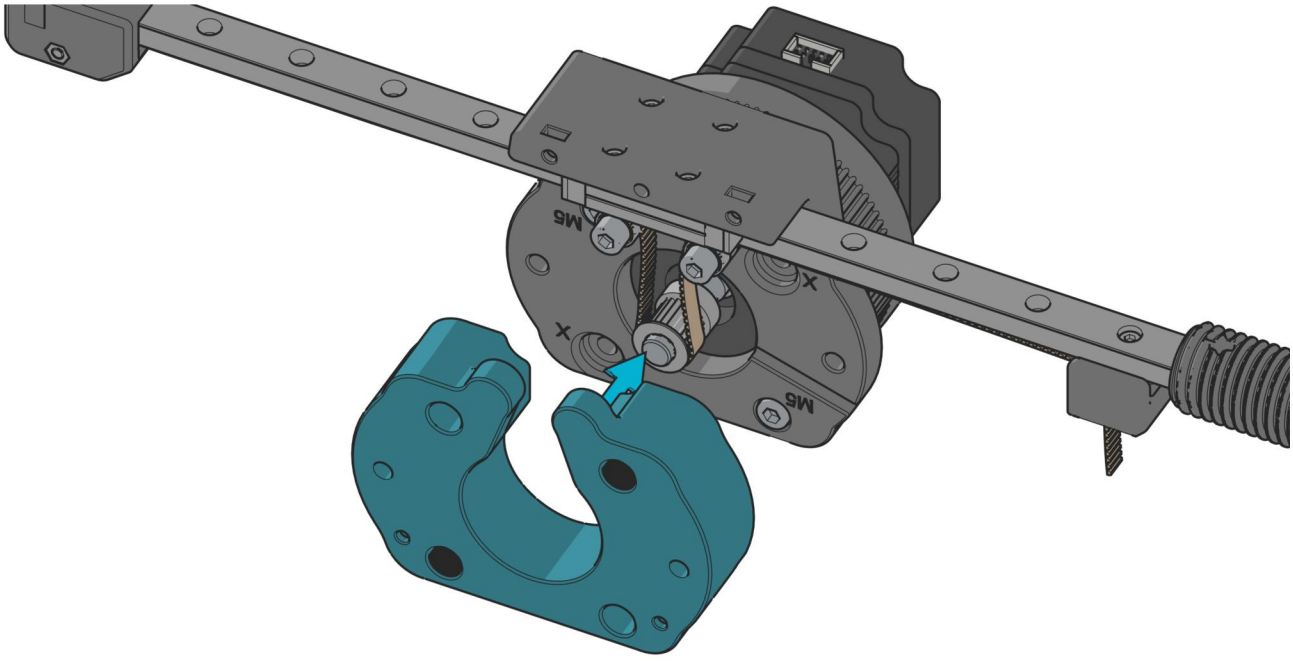
2×
M5×35 Screw

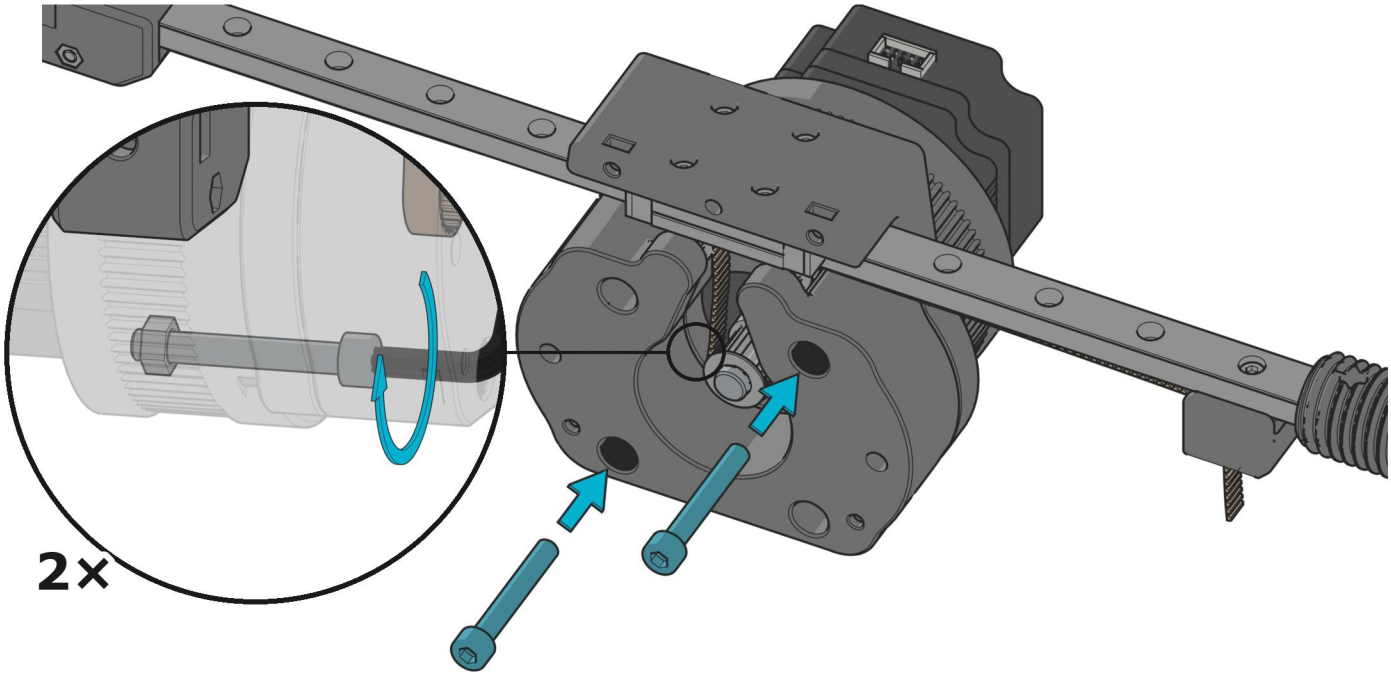


1×
Body middle



4 mm Hex key



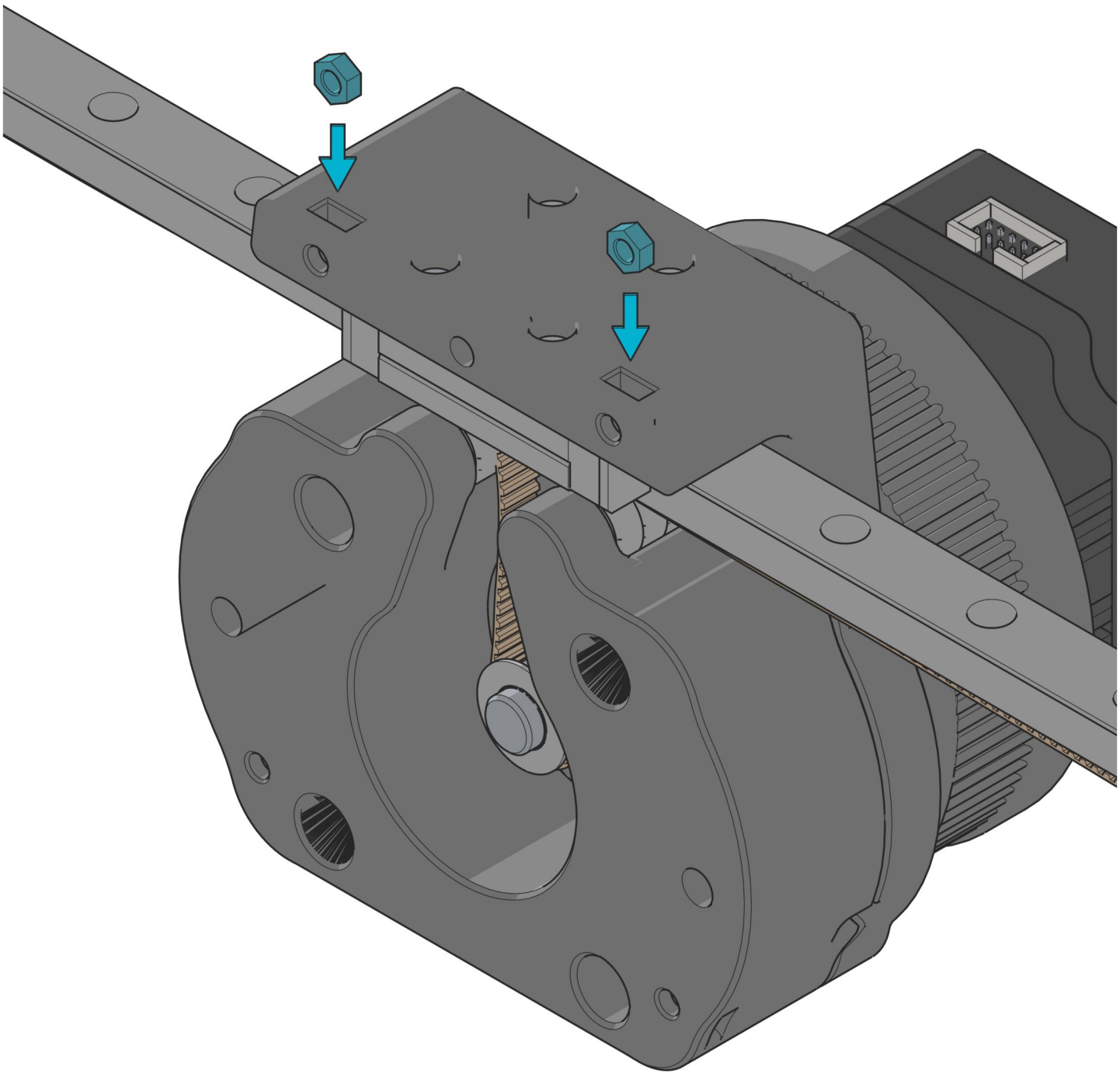


27



2×

M3 Nut

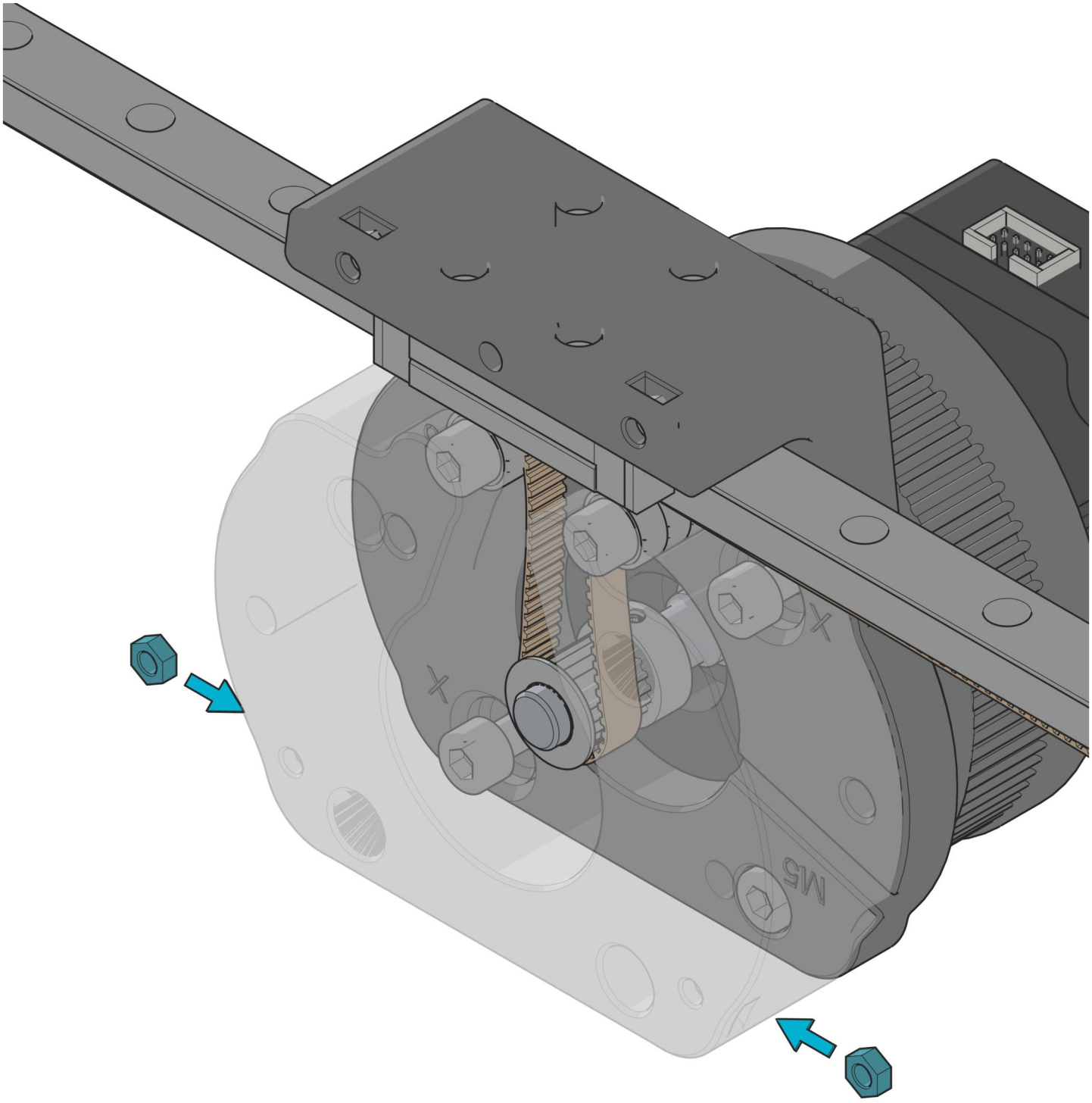


28



2×

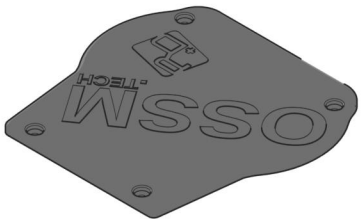
M3 Nut



29



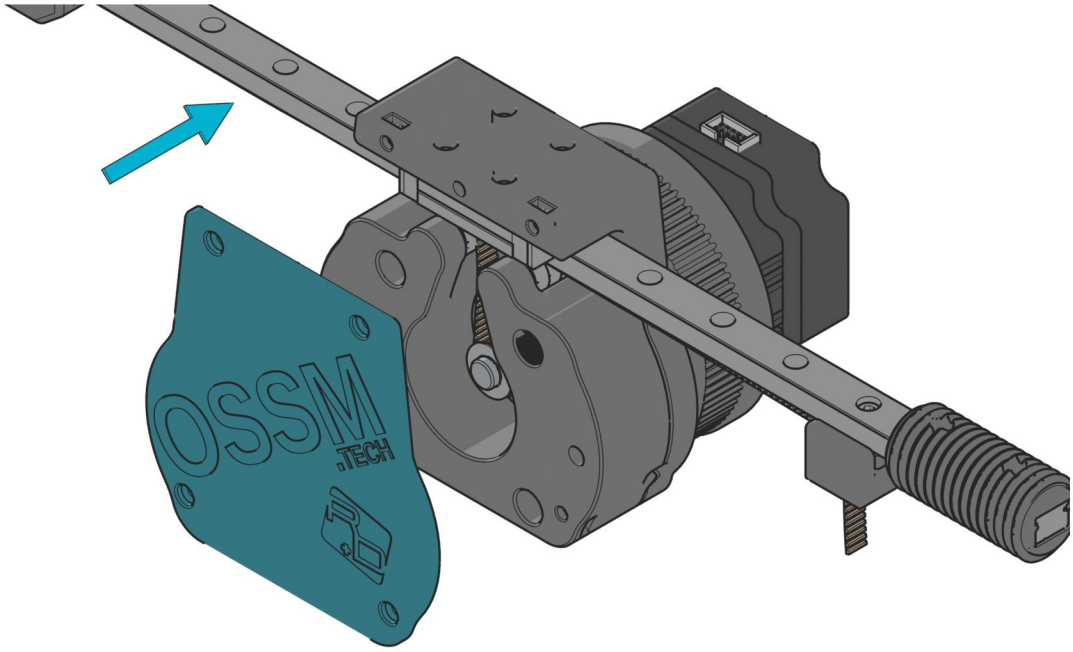
4×
M3×8 Screw

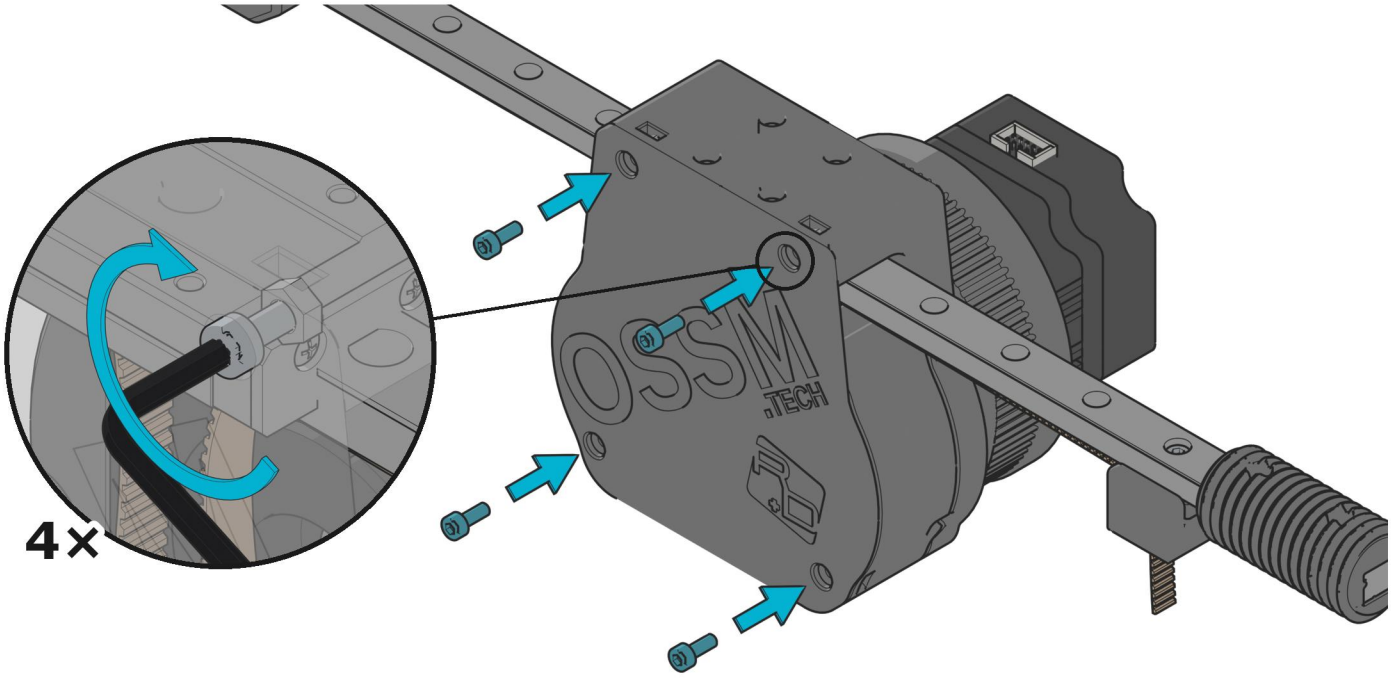


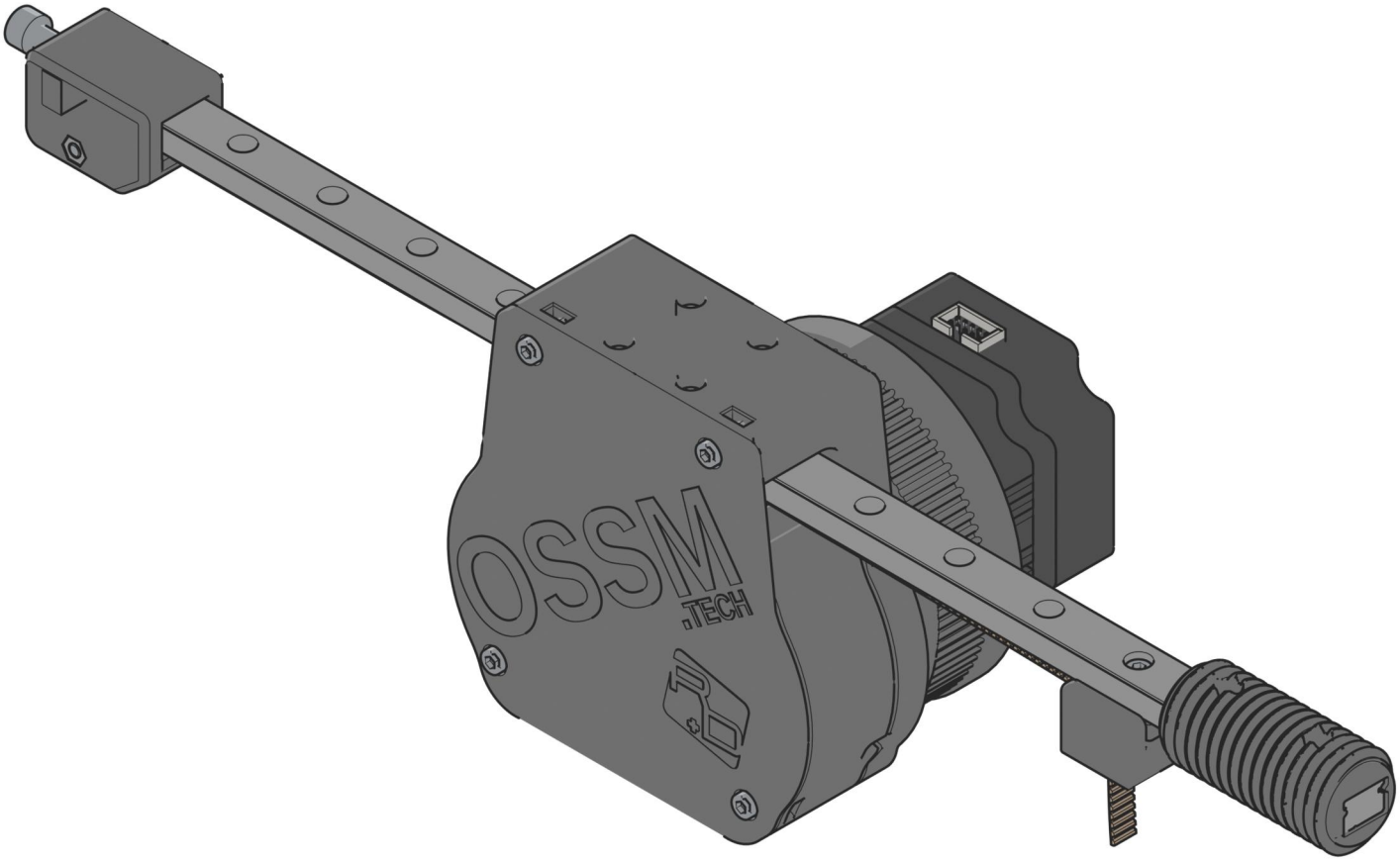
1×
Cover



2.5 mm Hex key







You may now proceed to the electronics and wiring.

RESEARCH + DESIRE

OSSM Actuator Build Guide

Actuator · Version 1.0.15

Support
support@researchanddesire.com
Assembly instructions
ohai.researchanddesire.com/ossm

Gloved hand: "Low Poly Gloved Hand" by CitrusH (@venture.dalton), CC BY 4.0 (creativecommons.org/licenses/by/4.0), posed and restyled: sketchfab.com/3d-models/low-poly-gloved-hand-1f5b60b45c4a43a0984042e23ced7fc6